

# Oil Spill Detection in PolSAR Imagery Using Composite Scattering Power Entropy and Multi-Scale Hybrid Feature Fusion Network

Deliang Xiang, Yu Lu, Dongdong Guan, Guannan Li, Jianda Cheng and Bangjie Li

**Abstract**—The issue of marine oil spills has been receiving increasing attention. Polarimetric Synthetic Aperture Radar (PolSAR) is effective in detecting oil spills by distinguishing backscatter differences between seawater and oil slicks. However, traditional polarimetric SAR features struggle to accurately distinguish between oil slicks and oil slick analogues generated by natural phenomena. In addition, using polarimetric features for oil spill detection faces challenges of poor performance in detecting elongated and small-scale oil spills. To address the interference from look-alikes, we introduce a composite polarimetric feature, termed composite polarimetric scattering entropy, which effectively differentiates between oil spills and look-alikes. To refine oil-water boundary segmentation, we propose an oil spill detection framework based on a Multi-Scale Hybrid Feature Fusion Network (MSHFFNet). The proposed method combines composite polarimetric scattering entropy with polarimetric intensity information and employs both square and strip pooling in the hybrid pooling module to capture a larger receptive field. This approach effectively captures information that helps distinguish elongated oil slicks. Additionally, in the feature fusion module, the method efficiently utilizes shallow features that contain detailed information, such as edges and textures, as well as deep features that contain semantic information. This significantly enhances the detection accuracy of small-scale oil slicks. Experiments conducted on Radarsat-2 data demonstrate that our method exhibits outstanding performance in reducing similar interferences and improving boundary accuracy, surpassing four state-of-the-art methods in terms of detection accuracy.

**Index Terms**—Polarimetric Synthetic Aperture Radar (PolSAR), Oil spill detection, Composite Polarimetric Scattering Power Entropy, Multi-Scale Hybrid Feature Fusion Network (MSHFFNet)

## I. INTRODUCTION

WITH the continuous rise in global energy demand, offshore oil extraction and transportation activities are becoming increasingly frequent. However, the growth in oil demand has led to frequent occurrences of marine oil spills. Tanker collisions, drilling equipment failures, subsea pipeline

leaks, and accidents triggered by natural disasters can all result in crude oil being spilled into the ocean. These incidents not only cause long-term damage to marine ecosystems but also have profound effects on the economies and livelihoods of regions that depend on marine resources [1]. In order to reduce the economic and environmental losses caused by marine oil spills, timely and accurate detection and delineation of spilled areas are crucial. Given this need, remote sensing technology, especially Synthetic Aperture Radar (SAR), with its almost all-weather capability, all-time, and active imaging capabilities, has become a powerful tool for monitoring marine oil spills [2]–[9].

In the observation of a single-polarization SAR system, oil slick attenuates the capillary waves and short gravity waves on the sea surface, reducing its roughness. Consequently, areas covered by oil films exhibit reduced backscattering signals to the sensor, resulting in significantly darker spots on the SAR images [10]–[12]. However, dark spots on SAR images are not solely caused by oil spills. Ship wakes, natural biogenic oils, low wind areas, and internal waves can also produce dark spots of low backscatter similar to oil spills, known as “look-alikes” [11]–[13]. The presence of these phenomena makes it challenging for single-polarization SAR to distinguish between oil spill areas and similar non-oil phenomena, leading to potential misclassifications. This limitation reduces the efficacy of single-polarization SAR in oil spill identification and detection [14]–[16].

The current SAR systems have evolved from the initial single-polarization SAR to the more advanced polarimetric SAR, which is divided into dual-polarization SAR and full-polarization SAR. Compared to dual-polarization SAR, full-polarization SAR can obtain more comprehensive scattering information, which reveals the different scattering characteristics between oil slicks and look-alikes [17]–[21]. When using full-polarization SAR data for oil spill detection, it is crucial to extract polarimetric features that can effectively distinguish between oil spill and look-alikes, and to construct a detection network that makes efficient use of these features.

The extraction of effective polarimetric features that can distinguish between oil films and look-alikes is the core task in full-polarization SAR oil spill detection. Skrunes *et al.* [19], [22] proposed an improved anisotropy  $A_{12}$ , based on the calculation of the two largest eigenvalues of the scattering matrix. They validated this method by using the Radarsat-2 dataset, demonstrating that in simple oil spill environments, the real part of the co-polarized cross-product and the geometric

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intensity can differentiate between plant and crude oil slicks. However, in more complex oil spill environments, due to the influence of other interfering factors on the sea surface, this method does not effectively distinguish between plant and crude oil slicks. Singha *et al.* [23] explored ten polarimetric parameters using full-polarization images from Radarsat-2 and TerraSAR-X, affirming the total polarized power *span* as a pivotal feature for detecting oil spill. Fan *et al.* [24] proposed the Bragg energy ratio  $\eta$ , a polarimetric parameter based on Freeman's three-component decomposition, which quantifies the relative contribution of Bragg scattering to the total power. This parameter was proven effective for oil spill detection on the SIR-C dataset. Liu *et al.* [25] developed a composite parameter,  $F$ , which integrates polarimetric entropy ( $H$ ), anisotropy ( $A$ ), average scattering angle ( $\alpha$ ), and correlation coefficient ( $\rho_{co}$ ). While this parameter showed promising results in detecting oil slicks, its performance at oil film edges is not as robust as the *span*. However, due to the normalization of  $H$ ,  $A$ ,  $\rho_{co}$  and  $\alpha$  in the combination features, the representation may tend to be biased towards one of these features. Tong *et al.* [26] refined and applied the self-similarity parameter  $rrr_s$ , utilizing it with other features through a random forest algorithm for oil spill detection. Nevertheless, this approach occasionally misclassifies the inner areas and boundaries of oil slick as look-alikes, leading to errors in oil region delineation.

In addition to selecting appropriate polarimetric features, developing effective detection methods is also crucial for achieving accurate oil spill detection. Skrunes *et al.* [22] noted that a single scattering feature is insufficient to fully describe the scattering differences between oil spills and the sea water background, and that detection methods relying on a single scattering feature limit the accuracy of detection. Consequently, numerous studies have attempted to identify and detect oil spills by combining polarimetric parameters and statistical methods. Li *et al.* [27] categorized polarimetric features into four types, including backscattered energy types, backscattering mechanism types, the correlation between polarization channels types, and a combination of polarimetric features. They input these four types of features into a random forest network to classify different thicknesses and types of oil films and evaluated the effectiveness of the features with quantitative metrics. However, traditional machine learning methods are unable to extract deep features. Relying only on shallow features inevitably introduces sea clutter in the detection results, which affects the accuracy of the detection. With the advancement of neural networks, deep learning has garnered widespread attention in the field of marine oil spill detection. Song *et al.* [28] designed a network utilizing Convolutional Neural Networks (CNN) to extract deep features, after which the deep features were fused and passed through RBF-SVM for oil spill detection. However, this method performs poorly in detecting small-scale oil spills, making it prone to misclassifying them as look-alikes. Subsequently, Song *et al.* [29] proposed a semantic segmentation model for marine oil spill detection. This method selects ten polarimetric features that effectively differentiate oil spills, using a random forest-based approach, and combines these features with spatial image characteristics for training.

However, this method has a high time complexity in large-scale scenarios and performs poorly in detecting elongated oil spills. Zhang *et al.* [30] designed a superpixel-based semantic segmentation model for oil spill detection, which incorporates superpixels as auxiliary information along with polarimetric data during network training. However, this method still has limitations in adapting to complex backgrounds. Wang *et al.* [31] introduced a semantic segmentation model combining Deeplabv3+ with a SVM classifier for oil spill detection. Although this method has achieved some success in binary classification problems, it produces incomplete segmentation results when dealing with elongated oil spill regions, making it prone to missed detection. In summary, large-scale SAR oil spill detection is a complex problem mainly due to the similarity between look-alikes and oil spills in polarimetric SAR images, leading to a high false detection rate. Meanwhile, small-scale and elongated oil spills pose additional challenges due to their small size and complex shapes, making them difficult to detect effectively using traditional segmentation methods. Therefore, developing detection methods that can comprehensively consider multiple polarimetric features and morphological characteristics is essential for improving the accuracy of oil spill detection.

To address these challenges in PolSAR oil spill detection, this paper proposes a method based on composite polarimetric scattering entropy and multi-scale hybrid feature fusion. The main contributions of this work are as follows: (1) we propose a feature, Composite Polarimetric Scattering Entropy ( $H_c$ ). Both qualitative and quantitative analyses demonstrate that  $H_c$  provides superior differentiation ability between look-alikes and oil slicks compared to other polarimetric features. (2) We have developed a semantic segmentation network for oil spill detection using polarimetric SAR data, which effectively enhances the ability to identify oil spill features by integrating composite polarimetric scattering entropy with polarimetric intensity information. Firstly, the network employs a hybrid pooling module to improve the detection of elongated oil spills, ensuring the accuracy of recognizing such specific shapes. Secondly, a feature fusion module is utilized to preserve the details and semantic information of polarimetric SAR images while reducing misclassification, thereby significantly improving the accuracy and robustness of the detection.

The remainder of this paper is structured as follows: Section II delineates the experimental data and methodologies employed. Section III elaborates on the settings of experimental parameters and provides a comparative analysis of various detection techniques. The principal discussions and conclusions of this study are expounded in Section IV and Sections V, respectively.

## II. DATASET AND METHOD

### A. Dataset

Fig. 1 presents three sets of Radarsat-2 fine-mode full-polarization SAR images utilized in this study. Fig. 1(a) shows an image from a controlled marine oil spill experiment conducted in the North Sea of Europe on June 8, 2011. This experiment released three different substances: crude oil,

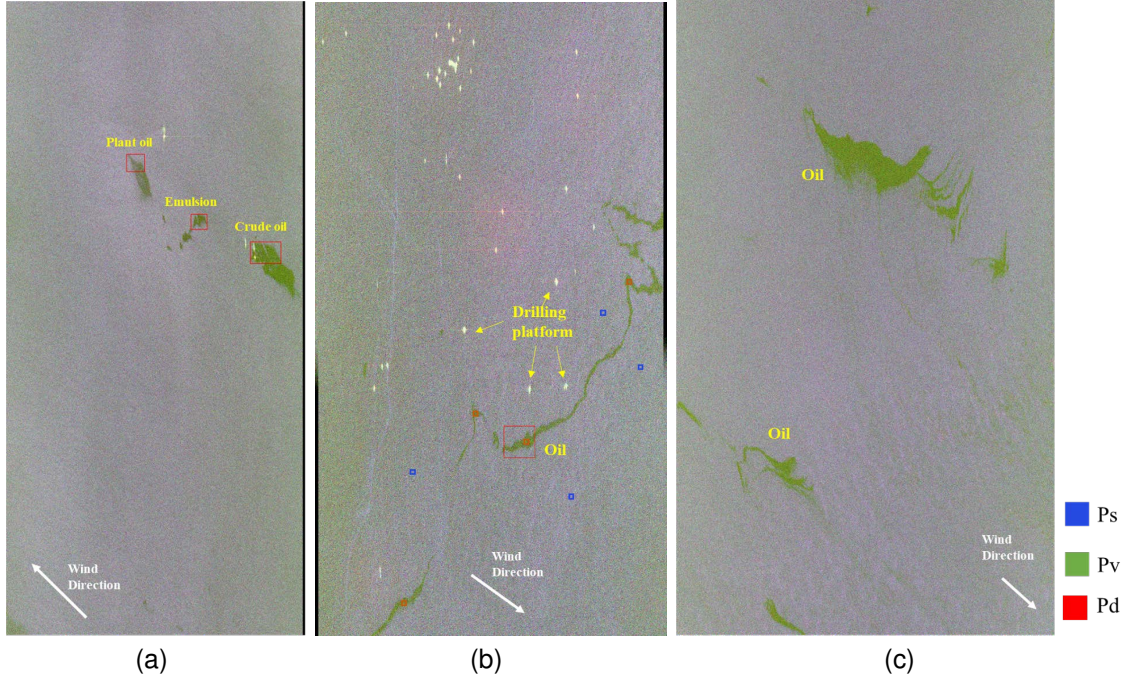


Fig. 1. Three Radarsat-2 fully polarimetric PauliRGB images used for the experiment: (a) European North Sea oil spill image, (b) Gulf of Mexico oil spill image I, (c) Gulf of Mexico oil spill image II.

emulsified oil and plant oil. Plant oil films, having surface-active compound structures similar to natural biogenic oil films, are commonly used as analogs for simulating natural biogenic oil films in experiments. Emulsified oil films, which are crude oil films containing water due to emulsification, were also considered as oil slicks in this experiment [19]. Fig. 1(b) and Fig. 1(c) show two different oil spill scenarios in the Gulf of Mexico on June 17, 2011, and May 8, 2010, respectively. Fig. 1(b) clearly depicts a long strip of oil slick resulting from a leak at an oil platform [28]. Fig. 1(c) represents frequent natural oil seepage in the area. Affected by wind, the oil slick is thicker on the leeward side than the windward side, leading to darker edges and creating a “feather-like” distribution of the oil film [27], [32], [33]. The image labels in this paper are determined based on ground truth from related literature [28], [34]. Subsequently, we invite oil spill detection experts to precisely adjust the labels, utilizing their extensive experience to ensure the accuracy of the labels. The detailed imaging parameters are outlined in Table I.

TABLE I  
INFORMATION ON RADARSAT-2 IMAGING USED FOR THE EXPERIMENT.

| Datasets        | European North Sea Oil Spill Image | Gulf of Mexico Oil Spill Image I | Gulf of Mexico Oil Spill Image II |
|-----------------|------------------------------------|----------------------------------|-----------------------------------|
| Radar Sensor    | Radarsat-2                         | Radarsat-2                       | Radarsat-2                        |
| Start Time      | 2011-06-08                         | 2011-06-17                       | 2010-05-08                        |
| SAR Band        | C                                  | C                                | C                                 |
| Beam Mode       | FQ15                               | FQ25                             | FQ23                              |
| Incidence Angle | 34.5-36.1E1(deg)                   | 43.6-44.9E1(deg)                 | 41.9-43.3E1(deg)                  |
| Polarimetric    | HH,VV,HV,VH                        | HH,VV,HV,VH                      | HH,VV,HV,VH                       |
| Wind Speed      | 1.6-3.3 m/s                        | 7.1m/s                           | 6.4m/s                            |

## B. Method

Fig. 2 presents the flowchart of the proposed method, which mainly consists of two parts: polarimetric feature extraction and a semantic segmentation model based on  $H_c$  and polarimetric scattering intensity information. Initially, the polarimetric SAR images are subjected to Lee filtering to mitigate the effects of speckle noise. This is followed by the extraction of polarimetric entropy  $H$ , Freeman-based polarimetric entropy  $H_p$ , and polarimetric intensity information from  $T_{11}$ ,  $T_{22}$  and  $T_{33}$ . The polarimetric feature  $H_c$  is formed by the simple product combination of  $H$  and  $H_p$ . Finally, this feature along with polarimetric scattering intensity information is used as inputs to the MSHFFNet, and the network is utilized to detect oil spill areas in various environmental conditions. Through quantitative evaluation analysis, the network is proven to accurately detect oil spills.

### 1) Describing Oil Spills Using Composite Polarimetric Power Entropy:

In full-polarization SAR, the scattering matrix  $S$  represents the complete information of the target scattering characteristics, which can be expressed as:

$$S = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}. \quad (1)$$

The matrix  $S$  is a symmetric matrix under the monostatic radar system, i.e.,  $S_{HV} = S_{VH}$ , it can be fully represented by the three-dimensional polarimetric coherence matrix  $T_3$  or the three-dimensional polarimetric covariance matrix  $C_3$ , defined

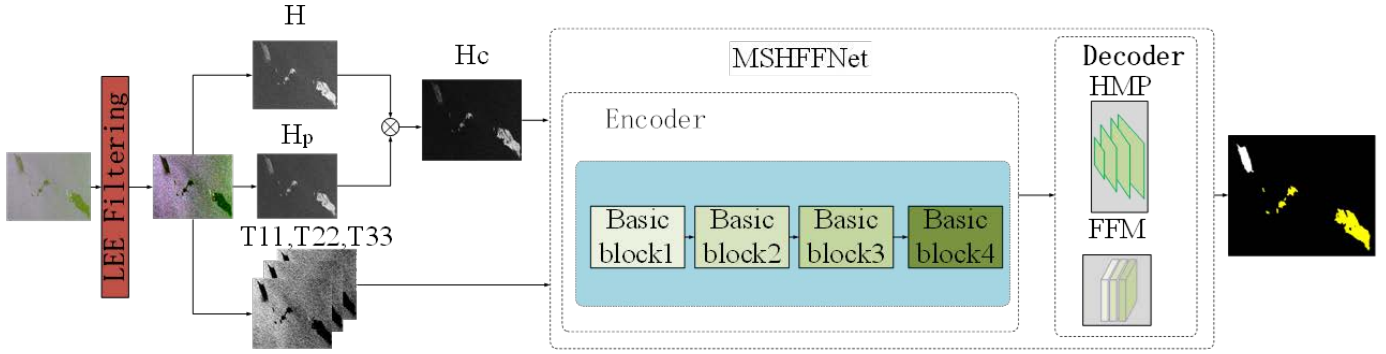


Fig. 2. Flowchart of the proposed method.

$$T_3 = \frac{1}{2} \begin{bmatrix} \langle |S_{HH} + S_{VV}|^2 \rangle & \langle (S_{HH} + S_{VV})(S_{HH} - S_{VV})^* \rangle & 2\langle (S_{HH} + S_{VV})S_{HV}^* \rangle \\ \langle (S_{HH} - S_{VV})(S_{HH} + S_{VV})^* \rangle & \langle |S_{HH} - S_{VV}|^2 \rangle & 2\langle (S_{HH} - S_{VV})S_{HV}^* \rangle \\ 2\langle S_{HV}(S_{HH} + S_{VV})^* \rangle & 2\langle S_{HV}(S_{HH} - S_{VV})^* \rangle & 4\langle |S_{HV}|^2 \rangle \end{bmatrix}. \quad (3)$$

as:

$$C_3 = \begin{bmatrix} \langle |S_{HH}|^2 \rangle & \sqrt{2}\langle S_{HH}S_{HV}^* \rangle & \langle S_{HH}S_{VV}^* \rangle \\ \sqrt{2}\langle S_{HV}S_{HH}^* \rangle & 2\langle |S_{HV}|^2 \rangle & \sqrt{2}\langle S_{HV}S_{VV}^* \rangle \\ \langle S_{VV}S_{HH}^* \rangle & \sqrt{2}\langle S_{VV}S_{HV}^* \rangle & \langle |S_{VV}|^2 \rangle \end{bmatrix} \quad (2)$$

$H_c$  proposed in this paper is primarily based on the component polarimetric entropy  $H$  from the Cloude decomposition. This describes the randomness of the scattering mechanism, defined as:

$$H = - \sum_{i=1}^3 P_i \log_3(P_i), \quad (4)$$

$$P_i = \frac{\lambda_i}{\sum_{j=1}^3 \lambda_j}, \quad (5)$$

where  $\lambda_i$  are the eigenvalues of the polarimetric coherence matrix  $T_3$ , and  $P_i$  are the pseudo-probabilities derived from the eigenvalues  $\lambda_i$ .

Inspired by polarimetric scattering entropy, Yang *et al.* [35] have proposed the scattering power entropy  $H_p$  based on Freeman's decomposition. It is defined as follows:

$$H_p = \sum_{i=1}^3 -P_i \log_3 P_i \quad 0 \leq H_p \leq 1, \quad (6)$$

where

$$P_1 = \frac{P_s}{P_s + P_v + P_d} \quad P_2 = \frac{P_v}{P_s + P_v + P_d} \quad P_3 = \frac{P_d}{P_s + P_v + P_d}. \quad (7)$$

$P_s$ ,  $P_v$  and  $P_d$  represent the powers of surface scattering, volume scattering and double-bounce scattering, respectively. Scattering entropy  $H_p$  is used to analyze and describe the type and complexity of scattering mechanisms. When  $H_p$  is low, scattering is primarily dominated by a single mechanism, whereas, a high  $H_p$  value indicates the presence of two to three predominant scattering mechanisms. Additionally, unlike  $H$ ,  $H_p$  does not exhibit roll-invariance.

In dealing with complex marine oil spill scenarios, it is crucial to utilize polarimetric features that can significantly

enhance the information on oil slick while minimizing the impact of look-alikes. Effectively combining the scattering characteristics of different polarimetric parameters, the combined features can significantly improve the discrimination between oil spills and other targets. In this paper, we combine polarimetric entropy and the polarimetric scattering power entropy based on Freeman's three-component decomposition into a composite polarimetric scattering power entropy  $H_c$ , thereby increasing the differentiation between seawater or look-alikes and oil slicks. It is defined as follows:

$$H_c = H \times H_p, \quad (8)$$

where  $H$  is the polarimetric entropy, and  $H_p$  is the polarimetric entropy based on Freeman's decomposition. Because  $H_p$  does not exhibit roll-invariance.  $H_c$  also does not have roll-invariance.

In the analysis of polarimetric features, the entropy value approaching 0 corresponds to a dominant scattering mechanism, such as a clean sea surface. In contrast, when multiple scattering occurs or the signal is disrupted by noise, the entropy value approaches 1, as seen in oil slick areas. [36] In look-alike scenarios, such as plant oil, the scattering mechanism remains Bragg scattering. The entropy value is slightly higher than that of a clean sea surface.

Meanwhile,  $H_p$  reflects the complexity of the scattering processes.  $H_p$  combines with the Freeman decomposition model to quantify the relative contributions of surface scattering, volume scattering, and double-bounce scattering mechanisms. If  $P_1$  is close to 1, it indicates a dominant scattering mechanism; if  $P_1 = P_2 = P_3$ , then  $H_p = 1$ , indicating the presence of all three scattering mechanisms. According to the sample in Fig. 3, as shown in Fig. 4(a), on a sea surface without oil slicks, the predominant scattering mechanism is surface scattering, characterized by low complexity. Consequently,  $H_p$  is close to 0 in Fig. 4(d). In contrast, as depicted in Fig. 4(b), on a sea surface covered with oil slicks, the proportions of surface scattering, volume scattering, and double-bounce scattering are equal, indicating higher complexity; hence,  $H_p$  is close to 1 in



Fig. 4(d). As shown in Fig. 4(c), the scattering mechanism of plant oil remains predominantly single scattering. Therefore, the  $H_p$  value falls between that of plant oil and seawater.  $H_p$  can be used to effectively distinguish oil spills from look-alikes.

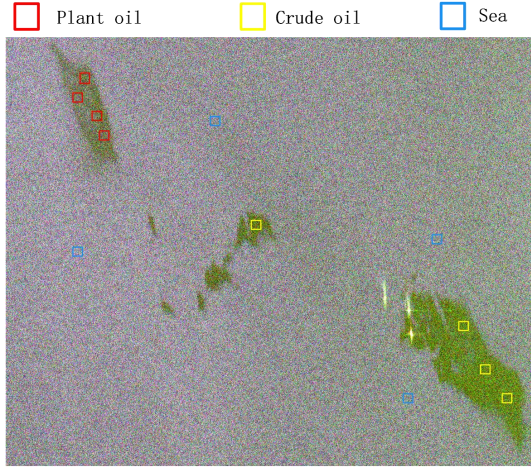


Fig. 3. Sample selection on European North Sea oil spill image.

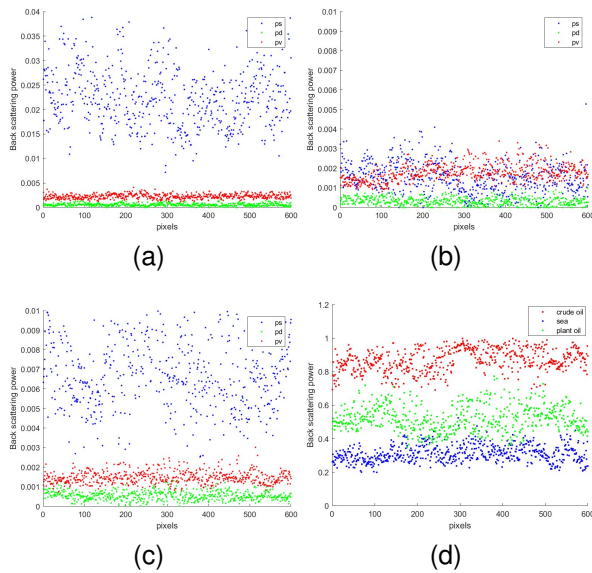


Fig. 4. Distribution of scattering mechanisms: (a) sea, (b) crude oil, (c) plant oil (d)  $H_p$

Therefore, to fully utilize the aforementioned two features, we introduce a polarimetric feature  $H_c$ , which integrates  $H$  and  $H_p$ . The value range of  $H_c$  is from 0 to 1. The combined feature represents the random scattering process. On clean sea surfaces, which are characterized by low complexity and minimal disorder in scattering mechanisms,  $H_c$  tends towards 0. Conversely, on oil-covered sea surfaces, where both the complexity and disorder of scattering mechanisms are elevated,  $H_c$  tends towards 1. By using a multiplicative form,  $H_c$  effectively increases the differences in polarimetric scattering characteristics between oil spills, seawater, and look-alikes, helping to suppress interference from look-alikes to oil slicks

and providing an effective polarimetric feature for oil spill detection. In order to visually demonstrate the advantages of the feature in distinguishing between crude oil, plant oil and seawater, Fig. 5 presents mean and variance plots for different regions based on the samples depicted in Fig. 3. From Fig. 5, it can be seen that  $H_c$  offers greater discriminative power between crude oil and plant oil compared to the other two features.

To more clearly illustrate the differences among the three features, we use the Gulf of Mexico oil spill image I from Fig. 1(b). As shown in Fig. 6(a),  $H$  shows a significant difference between the oil slick and seawater. However, in the regions of the oil platforms and oil slicks,  $H$  is close to 1. This indicates that relying solely on polarimetric entropy cannot effectively distinguish between the oil platforms and the oil slicks.  $H_p$  reflects the complexity of three physical scattering mechanisms: surface scattering, volume scattering, and double-bounce scattering. As shown in Fig. 6(b), the entropy difference between the oil slick and seawater in  $H_p$  is relatively small. The oil platform, dominated by a single scattering mechanism, has low complexity and is represented by low entropy in  $H_p$ . In contrast, the oil slick, with relatively balanced contributions from the three scattering mechanisms, exhibits higher scattering complexity and thus is represented by higher entropy. Fig. 6(c) demonstrates that the combined features, particularly  $H_c$ , effectively leverage the advantages of  $H$  and  $H_p$ , not only enabling the effective distinction between the oil slick and other interfering objects but also significantly enhancing the difference between seawater and the oil slick.

## 2) Evaluating the Impact of Fusion Feature Parameters on Oil Spill Identification Efficiency:

In this paper, we employ both qualitative and quantitative methods to evaluate the performance of the composite polarimetric scattering power entropy compared to other polarimetric features in distinguishing oil slicks, look-alikes and seawater. The quantitative analysis is conducted by calculating the Jeffreys-Matusita (J-M) distance and by analyzing the overlap areas between different categories in the probability density function (PDF) graphs to demonstrate the discriminative power of the feature. Qualitative assessments leverage these PDF graphs and visualizations to demonstrate the feature's effectiveness in correctly identifying distinct categories, thereby proving its efficacy.

To validate the effectiveness of the feature proposed in this paper, we conduct experiments using oil spill images from the North Sea in Europe. Additionally, we selected  $H$ ,  $H_p$ , and several features that have been proven to perform well in oil spill detection, including  $R_{co}$  [19],  $span$  [23], and  $\eta$  [24], with detailed information provided in Table II.

TABLE II  
INFORMATION ON RADARSAT-2 IMAGING USED FOR THE EXPERIMENT.

| Feature  | Formula                                            |
|----------|----------------------------------------------------|
| $R_{CO}$ | $R_{CO} =  \Re(\langle S_{HH} S_{VV}^* \rangle) $  |
| $span$   | $span = T_{11} + T_{22} + T_{33}$                  |
| $\eta$   | $\eta = \frac{T_{11} +  T_{12} ^2 / T_{11}}{span}$ |

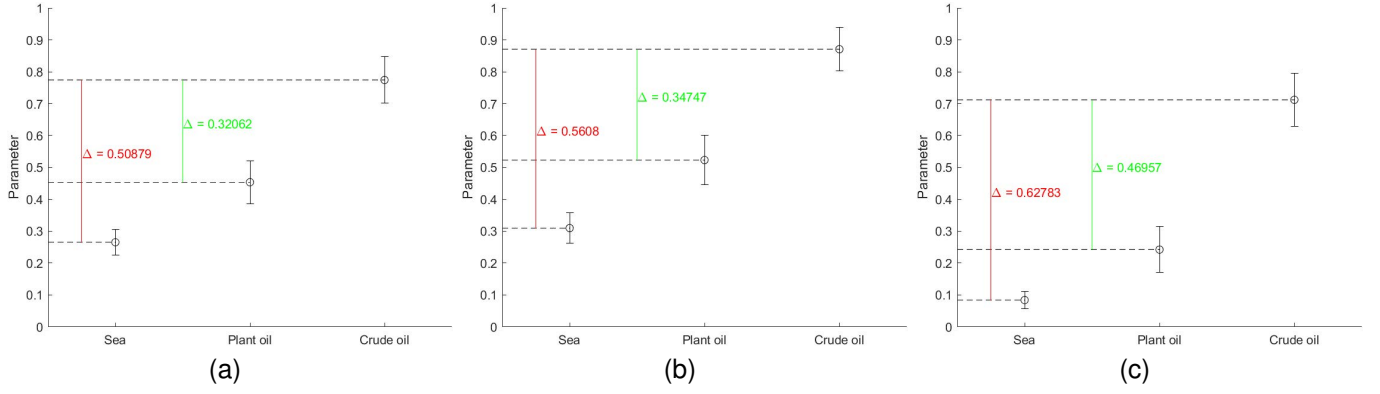


Fig. 5. Plots of mean and variance distributions for different features: (a)  $H$ , (b)  $H_p$ , (c)  $H_c$ .

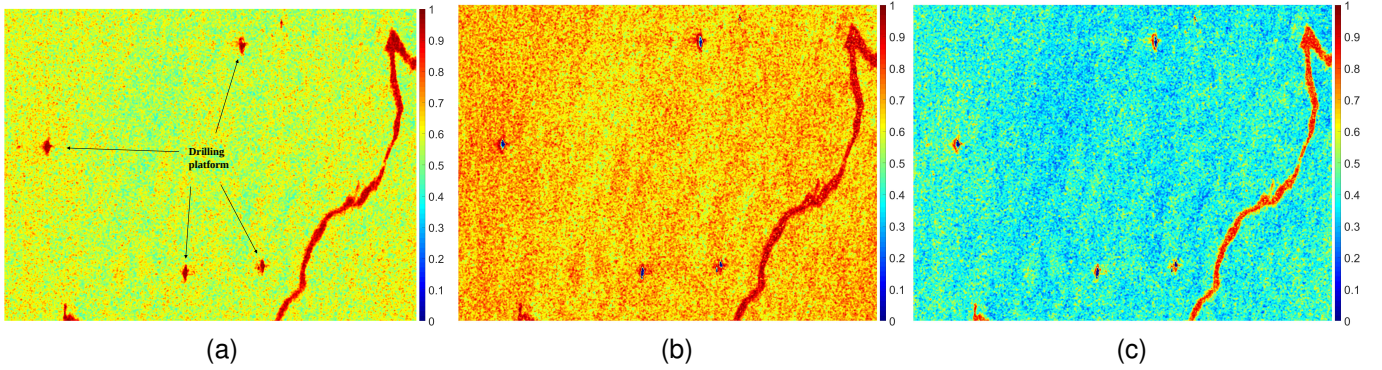


Fig. 6. Visualization of polarimetric features: (a)  $H$ , (b)  $H_p$ , (c)  $H_c$ .

To quantify the discriminatory capability of various polarimetric features between oil films and seawater, we employ the J-M distance for assessment. As an index measuring the separability between categories, the J-M distance is frequently used in feature selection due to its simplicity and broad applicability [37]. This index quantifies the distinction based on the distribution differences of two classes under a given feature. The formula for the J-M distance is as follows:

$$J = 2(1 - e^{-B}), \quad (9)$$

$$B = \frac{1}{8}(m_1 - m_2)^2 \frac{2}{\sigma_1^2 + \sigma_2^2} + \frac{1}{2} \ln \left( \frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1\sigma_2} \right), \quad (10)$$

where  $m_1$  and  $m_2$  represent the means of the two different targets, and  $\sigma_1^2$ ,  $\sigma_2^2$  are the variances of these targets, respectively. The J-M distance ranges from 0 to 2. The greater the separability between the distributions of oil slicks and seawater, the higher the J-M distance.

The results of the J-M distances for different categories are shown in Fig. 7. In the context of oil spill detection, the capability of features to distinguish between crude oil and seawater is the most critical consideration.  $span$ ,  $H$ ,  $H_p$  and  $H_c$  all exhibit J-M distances greater than 1.9 between crude oil and seawater, indicating their excellent performance in distinguishing these two classes. Furthermore, we also focus on the degree to which polarimetric features can distinguish between oil slicks and look-alikes. It can be observed that the J-M distance for  $H_c$  is the largest compared to other features.

Overall,  $H_c$  shows the largest J-M distances in distinguishing between crude oil and seawater, as well as between crude oil and plant oil. This indicates that this feature is effective in differentiating between oil slicks and seawater, and also significantly reduces misclassification with look-alikes.

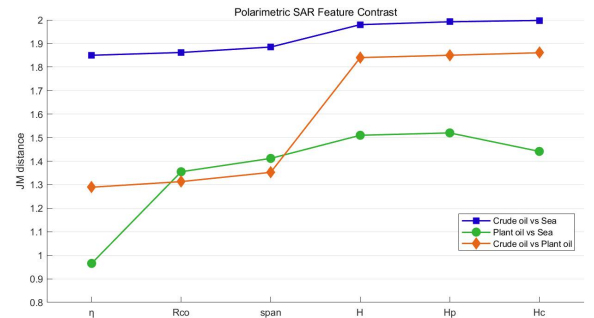


Fig. 7. J-M distance for polarimetric features.

In order to make the features comparable, based on the normalization and transformation of the features, the visualization results of  $H_c$  and other polarimetric features are shown in Fig. 8. In Fig. 8(a),  $\log R_{CO}$ , despite providing some differentiation between crude oil and seawater, fails to deliver a compelling visual contrast. As shown in Fig. 8(b), compared to  $\log R_{CO}$ ,  $1 - \eta$  exhibits significant visual differences between crude oil in the bottom right corner and seawater. However, the visual differences between crude oil in the center of the image and



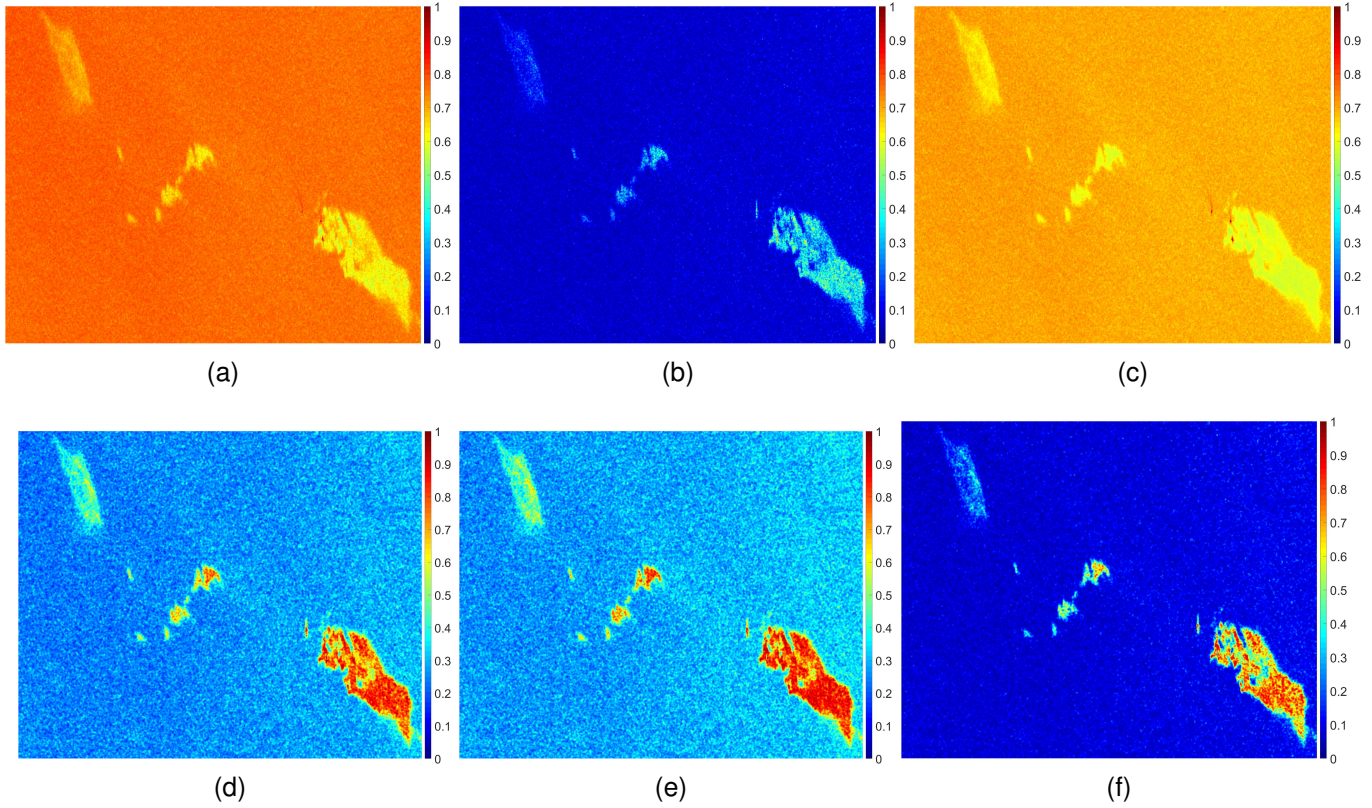


Fig. 8. Plot of visualization results for polarimetric features. (a)  $\log R_{CO}$ , (b)  $1 - \eta$ , (c)  $\log span$ , (d)  $H$ , (e)  $H_p$ , (f)  $H_c$ .

seawater are not pronounced. Fig. 8(c) shows that  $\log span$  lacks clear visual contrast between crude oil and plant oil. In Fig. 8(d)-(f), although  $H$ ,  $H_p$ , and  $H_c$  all visually differentiate between crude oil and plant oil, the visual contrast between seawater and the oil slick is more pronounced in  $H_c$  compared to  $H$  and  $H_p$ , with a more evenly distributed color palette in the seawater background. This indicates that  $H_c$  not only minimizes interference from sea clutter but also amplifies the signal of oil spills, thereby more distinctly identifying the differences between oil slicks and look-alikes.

The PDF graph serves as a qualitative assessment tool, enabling the analysis and visualization of polarimetric feature parameters' efficacy in differentiating between oil slicks and non-oil targets. The horizontal axis of this graph represents the feature values, while the vertical axis represents the proportion of pixels with respective feature values relative to the total number of pixels. The PDF graph displays the probability distribution of oil slicks and non-oil targets within a one-dimensional feature space. By analyzing the distance between different category probability density peaks and the size of the overlap area in the cross-section, we can determine the appropriate polarimetric features for distinguishing oil spill information from look-alikes.

As shown in the PDF graphs in Fig. 9, the distributions of crude oil, plant oil and seawater are depicted. Fig. 10 displays the overlap area sizes between different categories. Following the selection of samples and normalization of features, it is evident that under consistent sample conditions, the peak distances between seawater and crude oil in Figs. 9(d), 9(e),

and 9(f) are significantly smaller than those in Figs. 9(a), 9(b), and 9(c). In Fig. 10(a), compared to other feature parameters,  $H_c$  generally has a lower percentage of overlap between plant oil and crude oil. Moreover, in Fig. 10(b), the overlap percentage between crude oil and seawater is the lowest for  $H_c$ . Overall,  $H_c$  demonstrates the best performance in differentiating between oil slick and non-oil targets. This feature not only offers the greatest peak separation but also maintains a lower proportion of overlapping areas between different categories, thus providing a broader range of feature parameter choices for oil spill detection.

### 3) MSHFFNet:

In this paper, we introduce a PolSAR image oil spill detection method. The overall workflow is illustrated in Fig. 11. The MSHFFNet utilizes a mainstream encoder-decoder architecture that accommodates inputs of any size. In the encoder, the feature  $H_c$  and polarimetric scattering intensity information from  $T_{11}$ ,  $T_{22}$  and  $T_{33}$  are used as network inputs. Specifically, polarimetric scattering intensity information is chosen as a key input because it provides comprehensive intensity data, which has been proven to be directly or indirectly applicable to PolSAR image classification [28], [38], [39]. The encoder extracts features to obtain shallow features  $F_1$  and  $F_2$ , which contain detailed information, and deep features  $F_3$  and  $F_4$ , which contain semantic information. The decoder is composed of a feature fusion module and a hybrid pooling module. In the hybrid pooling module, the deep feature  $F_4$  is subjected to diverse pooling window sizes, culminating in a fusion of multiple features into the sophisticated feature.

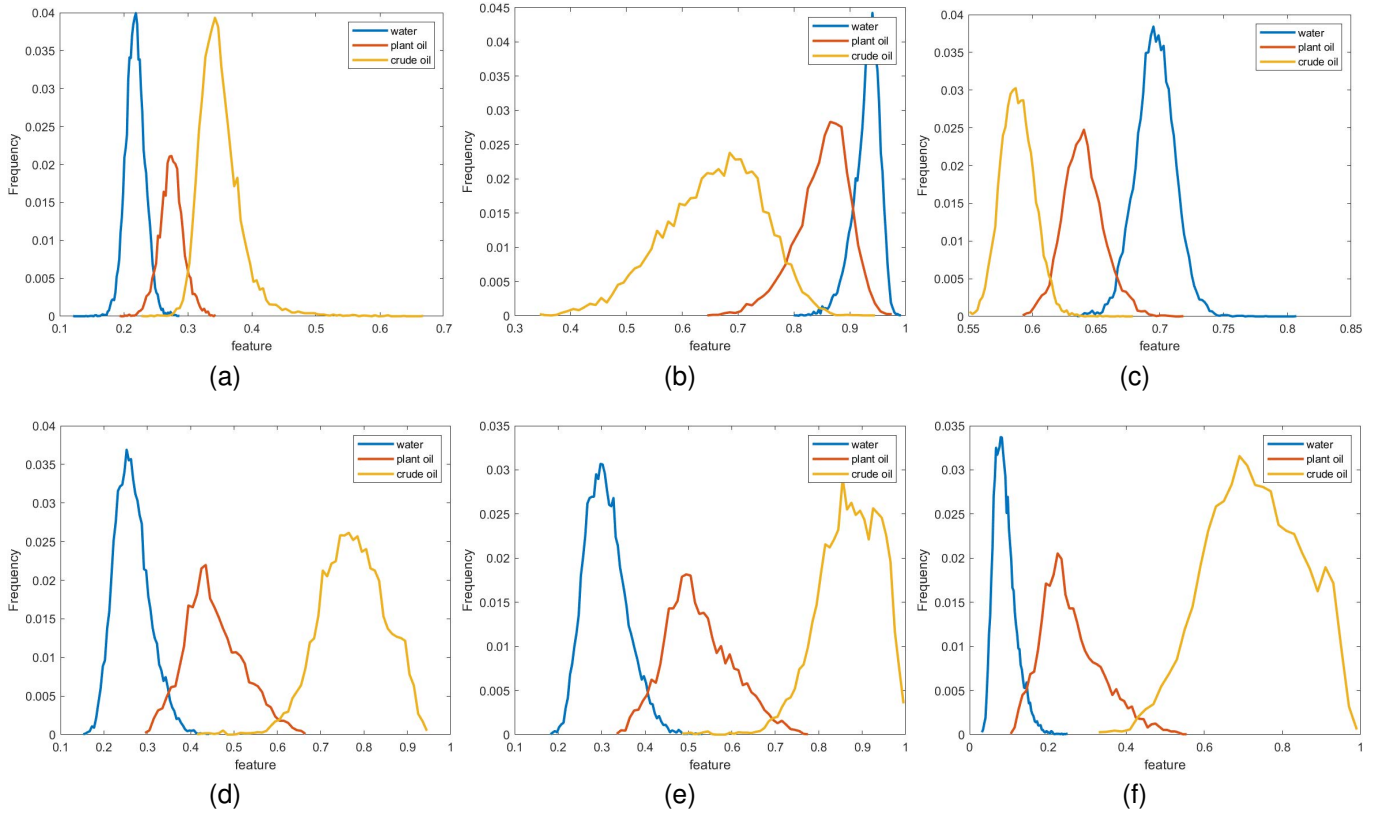


Fig. 9. Plot of probability density functions of polarimetric features in selected regions. (a)  $R_{CO}$ , (b)  $\eta$ , (c)  $span$ , (d)  $H$ , (e)  $H_p$ , (f)  $H_c$ .

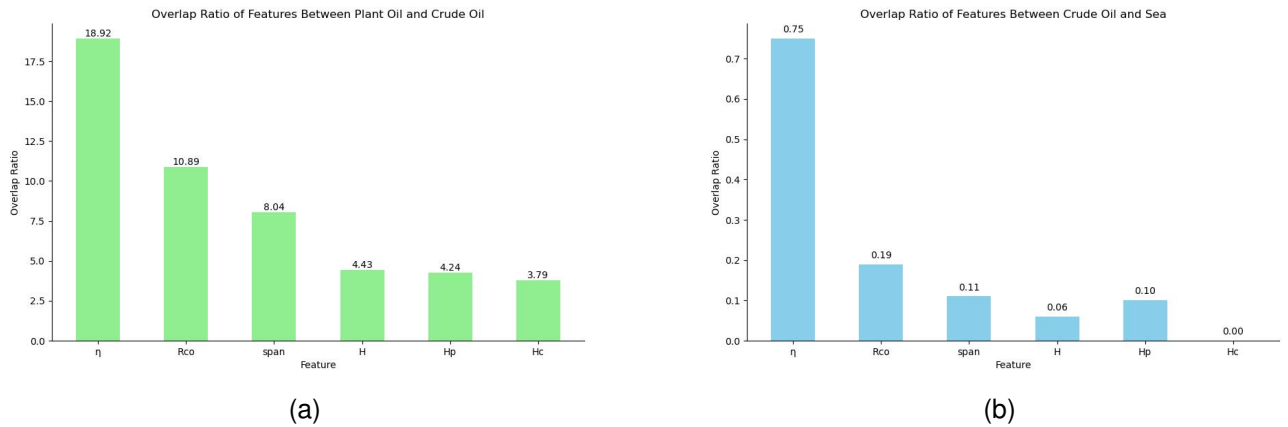


Fig. 10. Overlap area plots for different features. (a) Plant oil vs crude oil, (b) Crude oil vs Sea.

In the feature fusion module, shallow features  $F_1$ ,  $F_2$  and deep feature  $F_3$  are fused. During the fusion process, channel squeeze and excitation blocks (sSE) are used to weight the features, effectively integrating them. Subsequently, the fused features are further combined with the sophisticated feature of the hybrid pooling module to provide rich contextual spatial information. Finally, the segmented results are obtained through upsampling. The detailed network parameters are presented in Table III.

In the encoder, we select ResNet34 as the backbone network for feature extraction, with a downsampling rate set at 16

times. The choice of ResNet34 as the backbone is based on several key considerations. First, for polarimetric SAR images that contain rich polarization information, ResNet34 is sufficiently effective for feature extraction without the need for a deeper network structure. Secondly, the relatively moderate depth of ResNet34 facilitates easier training and reduced computational demand, potentially enhancing the speed of model convergence. When selecting the backbone network, we mainly considered the balance between feature extraction capability, computational efficiency, and generalization ability. For example, feature extraction structures with fewer layers,



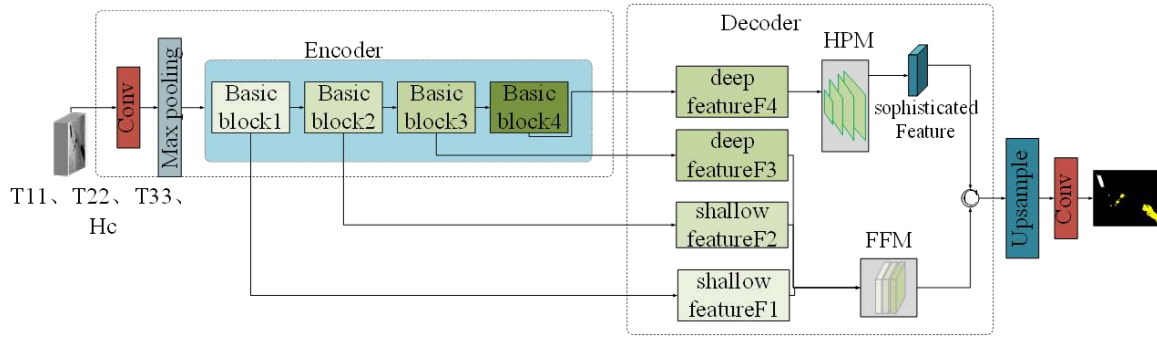


Fig. 11. Architecture of MSHFFNet.

TABLE III  
SPLIT NETWORK DETAILED PARAMETERS.

| blocks      | layers                       | output size                |
|-------------|------------------------------|----------------------------|
| Input Layer | input                        | $128 \times 128 \times 4$  |
| /           | <i>conv</i> ( $7 \times 7$ ) | $64 \times 64 \times 64$   |
| /           | max pooling                  | $32 \times 32 \times 64$   |
| resnet34    | basic block1                 | $32 \times 32 \times 64$   |
|             | basic block2                 | $16 \times 16 \times 128$  |
|             | basic block3                 | $8 \times 8 \times 256$    |
|             | basic block4                 | $8 \times 8 \times 512$    |
|             | AvgPool2d(2)                 | $2 \times 2 \times 512$    |
|             | <i>conv</i> ( $1 \times 1$ ) | $2 \times 2 \times 128$    |
|             | Upsample                     | $8 \times 8 \times 128$    |
|             | AvgPool2d(3)                 | $3 \times 3 \times 512$    |
|             | <i>conv</i> ( $1 \times 1$ ) | $3 \times 3 \times 128$    |
|             | Upsample                     | $8 \times 8 \times 128$    |
| HMP1        | AvgPool2d(4)                 | $4 \times 4 \times 512$    |
|             | <i>conv</i> ( $1 \times 1$ ) | $4 \times 4 \times 128$    |
|             | Upsample                     | $8 \times 8 \times 128$    |
|             | AvgPool2d(6)                 | $6 \times 6 \times 512$    |
|             | <i>conv</i> ( $1 \times 1$ ) | $6 \times 6 \times 128$    |
|             | Upsample                     | $8 \times 8 \times 128$    |
|             | <i>conv</i> ( $3 \times 3$ ) | $8 \times 8 \times 128$    |
|             | AvgPool2d(1,0)               | $1 \times 8 \times 512$    |
|             | <i>conv</i> ( $1 \times 3$ ) | $1 \times 8 \times 128$    |
|             | Upsample                     | $8 \times 8 \times 128$    |
| HMP2        | AvgPool2d(0,1)               | $8 \times 1 \times 512$    |
|             | <i>conv</i> ( $3 \times 1$ ) | $8 \times 1 \times 128$    |
|             | Upsample                     | $8 \times 8 \times 128$    |
|             | <i>conv</i> ( $3 \times 3$ ) | $8 \times 8 \times 128$    |
| FFM         | <i>conv</i> ( $3 \times 3$ ) | $8 \times 8 \times 128$    |
|             | Upsample                     | $16 \times 16 \times 256$  |
|             | sSE_Module                   | $16 \times 16 \times 256$  |
|             | <i>conv</i> ( $1 \times 1$ ) | $16 \times 16 \times 128$  |
|             | Upsample                     | $32 \times 32 \times 128$  |
|             | sSE_Module                   | $32 \times 32 \times 128$  |
|             | <i>conv</i> ( $1 \times 1$ ) | $32 \times 32 \times 64$   |
|             | sSE_Module                   | $32 \times 32 \times 64$   |
|             | <i>conv</i> ( $1 \times 1$ ) | $32 \times 32 \times 32$   |
|             | Upsample                     | $32 \times 32 \times 128$  |
| /           | <i>conv</i> ( $1 \times 1$ ) | $32 \times 32 \times 32$   |
| /           | Upsample                     | $128 \times 128 \times 32$ |
| /           | <i>conv</i> ( $3 \times 3$ ) | $128 \times 128 \times 3$  |

such as ResNet18, have faster computational speed but limited feature extraction capability due to the shallow depth, making it difficult to effectively identify fine edges of thin oil slicks and elongated oil patterns. On the other hand, deeper feature extraction structures, such as ResNet50, offer stronger feature extraction capabilities but significantly increase the number of parameters and computational cost. This not only raises the risk of overfitting when the training dataset is relatively small

but also demands higher computational resources. ResNet34, however, strikes a balance with its moderate network depth and residual structure, maintaining high feature extraction capabilities while controlling computational costs and avoiding overfitting. More importantly, the balance between feature representation ability and computational efficiency in ResNet34 enables it to achieve better generalization ability and stability when identifying complex patterns of oil slicks. Therefore, we chose ResNet34 as the network structure, which not only meets the requirements for feature extraction but also effectively prevents overfitting, ensuring the robustness of the model in practical applications. ResNet34 consists of four basic blocks, each containing 3, 4, 6 and 3 residual blocks (Resblocks) respectively. The features output by these four basic blocks are then fed into the feature fusion module and hybrid pooling module for further processing. The structure of the residual blocks is shown in Fig. 12.

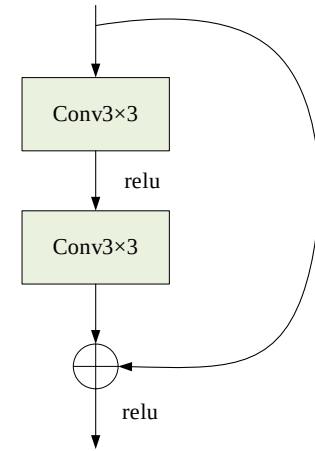


Fig. 12. Schematic diagram of the Resblock.

In the realm of PolSAR image classification, pyramid pooling is a method used to enhance the modeling capabilities of CNNs for long-range dependencies. The pyramid pooling module primarily collects contextual information based on standard spatial pooling operations, which acquire data from fixed square pools. However, this method may lack the necessary flexibility to adeptly handle complex oil spill scenarios. As illustrated in Fig. 13(a) and Fig. 13(c), square pooling has several disadvantages [40], [41]. For instance, on the ocean

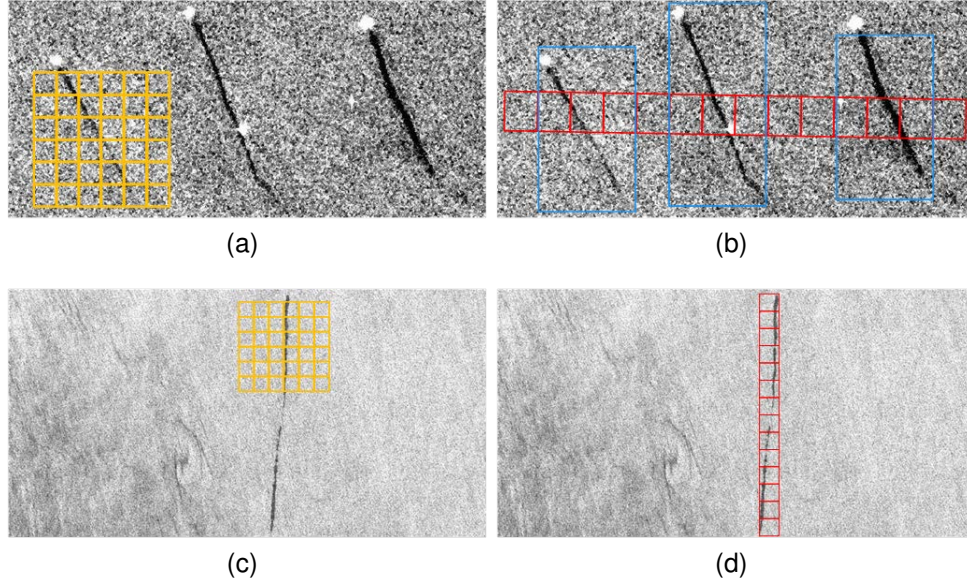


Fig. 13. Square pooling and strip pooling work differently. (a) Square pooling for obtaining information on discretely distributed oil slicks, (b) Strip pooling to obtain information on discretely distributed oil slicks, (c) Square pooling to get Information on long strips of oil slicks, (d) Strip pooling to get information on long bands of oil slicks.

surface, oil spills from tankers may appear as elongated streaks or be dispersed. The conventional square pooling approach can inadvertently capture extraneous information from adjacent areas, compounding the challenge. To more efficiently capture the long-distance dependencies of oil slicks, we introduce strip pooling [42]. The definitions for horizontal strip pooling  $y_i^h$  and vertical strip pooling  $y_j^v$  are as follows:

$$y_i^h = \frac{1}{W} \sum_{0 \leq j \leq W} x_{i,j}, y_j^v = \frac{1}{H} \sum_{0 \leq i \leq H} x_{i,j}, \quad (11)$$

where  $x$  is the input two-dimensional feature tensor, and  $H$ ,  $W$  are the dimensions of the tensor's height and width, respectively. Strip pooling averages all the feature values across a single row or column. As shown in Fig. 13(b), this method achieves distant connections in discrete oil spill regions by applying a longer kernel shape along one spatial dimension. Meanwhile, as shown in Fig. 13(d), it maintains a narrow kernel shape along another spatial dimension. This helps to compactly capture the local context and avoid interference from unrelated areas, thereby enhancing the accuracy of oil spill detection. The hybrid pooling module, by combining standard pooling and strip pooling, utilizes pooling operations with different kernel shapes to effectively capture both short-range and long-range dependencies in oil spills under complex environments.

The design of the hybrid pooling module introduced in this paper is depicted in Fig. 14, which is structured into two distinct submodules. In the first submodule, deep features  $F_4$ , extracted by the encoder, are subject to spatial pooling with varying kernel sizes, producing feature maps of multiple scales. To fully perceive local context information in tightly packed oil spill areas, the pooling layers are designed with reasonable intervals, setting the kernel sizes to  $2 \times 2$ ,  $3 \times 3$ ,  $4 \times 4$  and  $6 \times 6$ . These feature maps are subsequently processed using

a " $1 \times 1$  Conv" to streamline the channel count. Subsequently, features are upsampled to match the size of the feature maps before the pyramid module and are then concatenated along the channel dimension.

In the second submodule, the deep feature  $F_4$  utilizes vertical and horizontal strip pooling operations to capture dispersed or elongated oil spill information. To further enhance feature utility and efficiency, these maps undergo compression via " $3 \times 1$  Conv" and " $1 \times 3$  Conv" operations. Similarly, upsampling is employed to restore the feature maps to their original dimensions prior to pooling. Finally, the feature maps produced by two submodules are concatenated along the channels with the pre-pooling features to yield the sophisticated feature.

To alleviate the difficulty of detecting small-scale oil slick regions, a feature fusion technique was designed. This technique combines feature information from different levels of deep neural networks. It utilizes the rich spatial detail information contained in shallow features, such as edges and textures, as well as the abstract semantic information from deep features. Such fusion of features ensures that detail information is preserved during the decoding process while also enhancing semantic information. Based on this, to highlight potential key areas containing oil spill information, such as tiny droplets or thin oil films, this method introduces the sSE module. The sSE block calculates the importance of spatial locations of the input features, reducing interference from non-oil areas and helping the model to more clearly differentiate oil spill areas. The mathematical formulation of the sSE block is presented as:

$$\hat{\mathbf{U}}_{sSE} = [\sigma(q_{1,1}) \mathbf{u}^{1,1}, \dots, \sigma(q_{i,j}) \mathbf{u}^{i,j}, \dots, \sigma(q_{H,W}) \mathbf{u}^{H,W}], \quad (12)$$

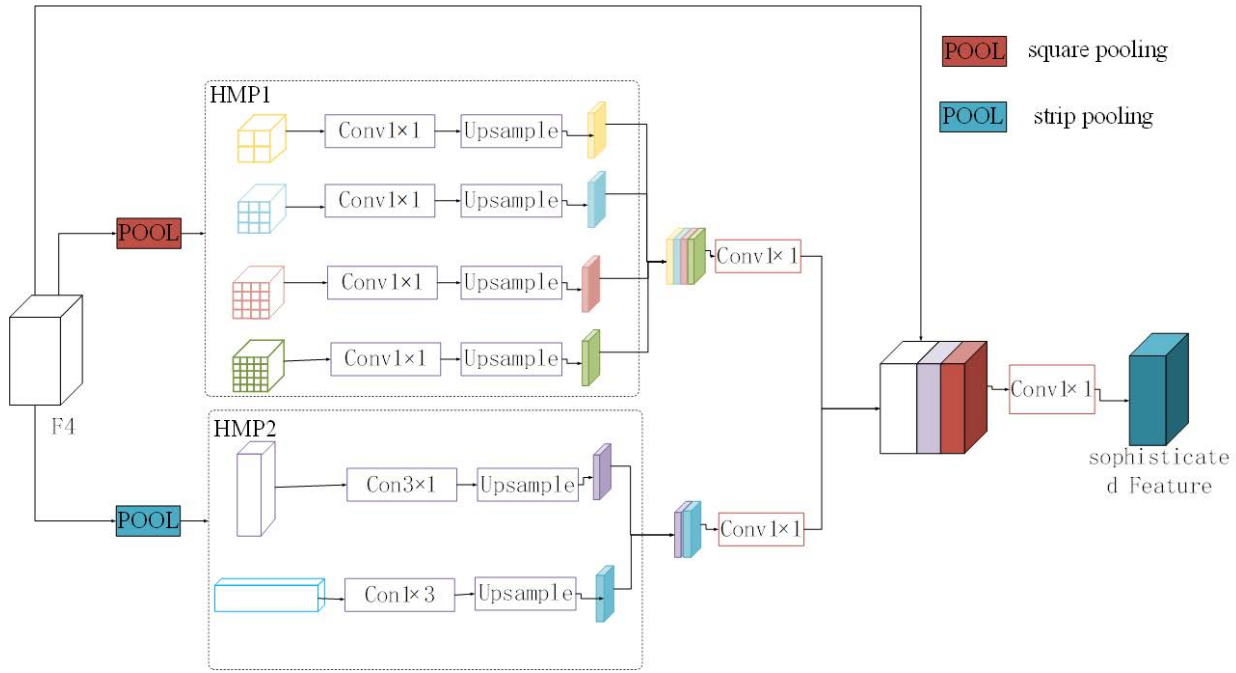


Fig. 14. Hybrid Pooling Module Diagram.

where  $\mathbf{U}$  is the input feature matrix,  $H$  and  $W$  are its height and width, respectively.  $\sigma$  represents the Sigmoid function,  $q_{i,j}$  are the importance scores for each spatial position calculated by the sSE module, and  $\hat{\mathbf{U}}_{sSE}$  is the feature map weighted by the sSE block. Fig. 15 illustrates the schematic of sSE block.

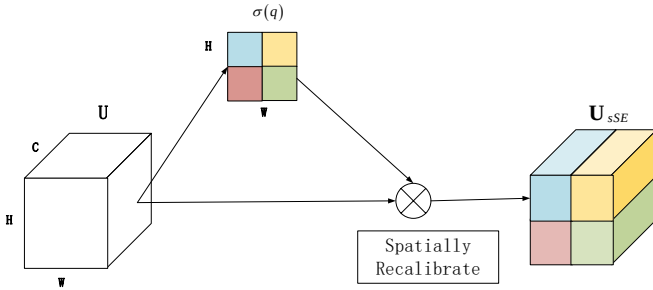


Fig. 15. Schematic diagram of channel squeeze and excitation blocks.

The model structure is depicted in Fig. 16. We utilize features extracted from the Resnet34 network as inputs for a multi-dimensional feature fusion module, which are designated as shallow feature  $F_1$ , shallow feature  $F_2$  and deep feature  $F_3$ . Initially, deep feature  $F_3$  is upsampled to align with the dimensions of shallow feature  $F_2$ . Subsequently, this feature is then refined using an sSE block, which emphasizes crucial information regarding oil slicks locations while mitigating background disturbances. The enhanced  $F_3$  is then concatenated with  $F_2$  along the channel dimension. To optimize the channel count, a  $1 \times 1$  Conv is employed for dimension reduction. Next, the fused feature map  $S_1$  is upsampled to match the resolution of  $F_1$  and further enhanced for targeted oil spill areas using the sSE block. After concatenating  $S_1$  with  $F_1$ , a  $1 \times 1$  Conv is used to reduce dimensions, resulting in

the fused feature map  $S_2$ . Finally,  $S_2$  is processed through the sSE block to produce the integrated feature  $S_3$ , highlighting oil spill regions in the feature map. This method utilizes the fusion of shallow features to capture detailed and edge information of oil slicks and enhances the targeted oil spill areas through weighted processing by the sSE block, thereby maintaining high precision and efficiency in handling oil spills of various scales.

In this paper, given the significantly smaller number of oil spill samples compared to seawater samples in the dataset, direct training could potentially lead to non-convergence of the model. To address this issue, we utilize a combined loss function of dice Loss and focal Loss to optimize the model. Dice Loss improves the model by comparing the similarity between the predicted and actual regions, focusing specifically on the model's ability to recognize minority classes. The formula for dice Loss is as follows:

$$L_{\text{Dice}} = 1 - \frac{2 \sum_{k=1}^M y_k \hat{y}_k}{\sum_{k=1}^M y_k + \sum_{k=1}^M \hat{y}_k}, \quad (13)$$

where  $y_k$  represents the label value of pixel  $k$ ,  $\hat{y}_k$  represents the predicted value of pixel  $k$ , and  $M$  is the total number of pixels. Focal Loss is designed to reduce the contribution of easy-to-classify samples to the total loss, focusing instead on those samples that are difficult to classify. It is defined as follows:

$$L_{\text{Focal}} = -\beta_t (1 - p_t)^\gamma \log(p_t), \quad (14)$$

where  $\gamma$  is a tunable parameter.  $\beta$  scales down the influence of easy-to-classify samples, and it typically ranges between 0 and 1.

The total loss function,  $L_{\text{Total}}$ , is defined as follows:

$$L_{\text{Total}} = L_{\text{Dice}} + L_{\text{Focal}}. \quad (15)$$



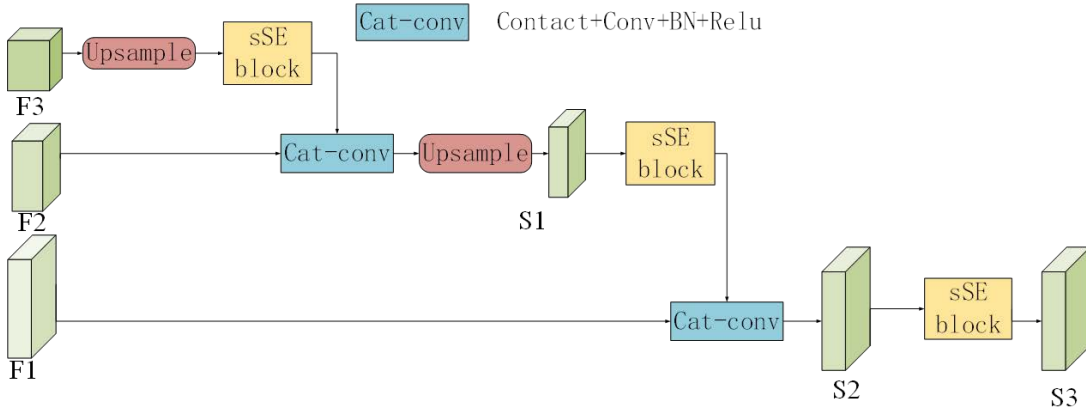


Fig. 16. Structure of feature fusion module.

Through a series of parameter tuning experiments, setting  $\beta = 0.75$ ,  $\gamma = 2$  in this experiment has proven to yield good results in oil spill detection.

#### 4) Evaluation Metrics:

In this experiment, a series of evaluation metrics are used to assess the performance of different models, including Overall Accuracy (OA), Average Accuracy (AA), F1 Score, and Mean Intersection over Union ( $mIoU$ ) as metrics to measure model performance. OA represents the proportion of pixels correctly identified as oil spills relative to the total number of pixels, defined as:

$$OA = \frac{TP + TN}{TP + TN + FP + FN}, \quad (16)$$

where  $TP$ ,  $TN$ ,  $FP$  and  $FN$  denote the number of true positives, true negatives, false positives, and false negatives, respectively.

AA represents the average of the accuracies for each category. It is defined as:

$$AA = \frac{1}{2} \times \left( \frac{TP}{TP + FN} + \frac{TN}{FP + TN} \right). \quad (17)$$

F1 is considered the harmonic mean of the model's precision and recall, defined as:

$$F1 = \frac{2 * TP}{2 * TP + FN + FP}. \quad (18)$$

$mIoU$  calculates the average of the Intersection over Union ( $IoU$ ) for each category, representing the ratio of the intersection to the union of the predicted and actual segmented areas, defined as:

$$mIoU = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i + FN_i}, \quad (19)$$

where  $i$  represents each category.

### III. EXPERIMENTAL RESULT

#### A. Experimental Setting

In Fig. 1(a) and Fig. 1(b), the regions within the red boxes are designated as the training set, and these areas are divided into  $128 \times 128$  sub-images. To prevent model overfitting caused

by a small dataset size, we applied geometric augmentation techniques such as  $90^\circ$  rotation and mirror flipping to enhance the model's generalization ability. After augmentation, a total of 596 patches of size  $128 \times 128$  were obtained for the training set. Since the network is designed to detect entire images, the regions outside the training set were used as the test set. Specifically, the pixel size of Fig. 1(a) is  $7120 \times 3369$ , and that of Fig. 1(b) is  $7043 \times 4096$ . In addition, Fig. 1(c), which was not used as the training set, with a pixel size of  $6660 \times 4096$ , was also included in the test set. In this paper, the network input channel count is 4, the optimizer used is SGD, and the experimental platform is based on a Linux system, with hardware and software configurations detailed in Table IV. To evaluate the performance of the proposed method, we compare it with four different PolSAR image oil spill detection methods, including RF [26], Improved-2D-CNN [28], Dual-EndNet [29] and DRSNet [31]. To ensure a fair comparison, all methods utilize the same training samples.

TABLE IV  
EXPERIMENTAL ENVIRONMENT PARAMETER SETTING.

| setting | vision                       |
|---------|------------------------------|
| CPU     | Intel(R) Xeon(R) Silver 4210 |
| GPU     | Nvidia GeForce RTX 4090      |
| Memory  | 64GB                         |
| Pytorch | 1.10.1                       |

#### B. Comparison and analysis of result

##### 1) Experimental results for European North Sea oil spill image:

To validate the proposed method on challenging PolSAR oil spill images, we conducted experiments on plant oil and crude oil. The quantitative evaluation metrics are shown in Table V. The method presented in this paper achieves higher detection accuracy for plant oil, crude oil and seawater than other methods on oil spill images from the North Sea in Europe. The OA value of our method is higher by 3.68%, 1.9%, 0.97% and 0.88% compared to RF, Improved-2D-CNN, Dual-EndNet and DRSNet, respectively. The AA value of our method is higher by 18.16%, 7.53%, 1.61% and 2.09% compared to RF,

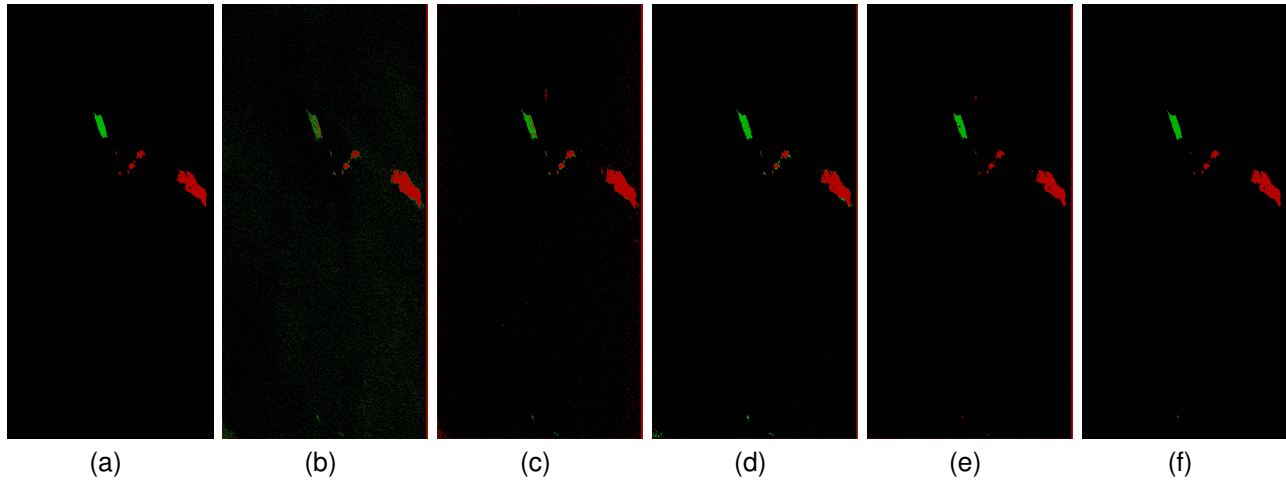


Fig. 17. Results of different methods for imagery of oil spills in the European North Sea. (a) Groud Truth, (b) RF, (c) Improved-2D-CNN, (d) Dual-EndNet, (e) DRSNet, (f) MSHFFNet.

TABLE V  
ACCURACY OF METHODS IN EUROPEAN NORTH SEA OIL SPILL IMAGE.

| Method          | Seawater     | Plant oil    | Crude oil    | $OA$         | $AA$         | $F1$        | $mIoU$       |
|-----------------|--------------|--------------|--------------|--------------|--------------|-------------|--------------|
| RF              | 96.37        | 51.48        | 91.53        | 96.26        | 79.80        | 0.60        | 54.70        |
| Improved-2D-CNN | 98.09        | 76.25        | 96.94        | 98.04        | 90.43        | 0.70        | 58.67        |
| Dual-endNet     | 99.01        | 96.18        | 93.84        | 98.97        | 96.35        | 0.85        | 76.88        |
| DRSNet          | 99.16        | 92.00        | 96.53        | 99.06        | 95.87        | 0.93        | 87.57        |
| MSHFFNet        | <b>99.96</b> | <b>96.52</b> | <b>97.40</b> | <b>99.94</b> | <b>97.96</b> | <b>0.97</b> | <b>94.04</b> |

Improved-2D-CNN, Dual-EndNet and DRSNet, respectively. Our method's  $F1$  and  $mIoU$  scores are 0.97 and 94.04, significantly outperforming RF, Improved-2D-CNN, Dual-EndNet and DRSNet. This proves that our method can more accurately identify and segment oil spill regions, reducing false positives and misses.

Fig. 17 shows the detection results of our algorithm and comparative experiments. As shown in Fig. 17(b), the noise in SAR images inevitably affects traditional methods. The detection result of RF has many speckles on the sea surface, likely due to traditional methods' inability to extract deep features. As shown in Fig. 17(c), compared to RF, the results of Improved-2D-CNN still have many noise-related issues affecting the detection outcome, possibly because CNNs rely on square domain perception of features without fully utilizing contextual information. As illustrated in Fig. 17(d) and Fig. 17(e), the detection results from Dual-EndNet and DRSNet have significantly fewer speckles on the sea surface, with errors mainly concentrated around the boundaries of crude oil. In Fig. 17(f), it is visually apparent that our method provides better detection of crude oil and plant oil. To more clearly demonstrate our proposed method's discriminative power between crude oil and plant oil, we zoom in on the oil spill area.

As shown in Fig. 18, the enlarged detection results for the oil spill area are presented. Fig. 18(b), Fig. 18(c) and Fig. 18(d) illustrate that RF, Improved-2D-CNN and Dual-EndNet mistakenly detect the edges of the two oil areas on the right as plant oil to varying degrees. This error may stem from

the edges of the crude oil being altered by weathering and emulsification processes, which induce scattering mechanisms similar to those of plant oil films. Furthermore, as shown in Fig. 18(e), compared to RF, Improved-2D-CNN and Dual-EndNet, the crude oil areas are better detected, but there are still misclassifications of the interiors of crude oil and plant oil as seawater. In Fig. 18(f), it is evident that our method achieves more accurate detection in small-scale oil slick regions, closely matching the ground truth and significantly reducing misclassification events between crude oil and vegetable oil films. This demonstrates that our method possesses exceptional capability in mitigating the interference of similar substances and accurately identifying small-scale oil slicks. The improvements introduced by this method not only enhance detection accuracy but also significantly improve the reliability of oil spill identification in complex marine environments.

## 2) Experimental results for Gulf of Mexico oil spill image I:

The oil spill region in image I of the Gulf of Mexico oil spill appears as elongated bands, and the experimental detection results are shown in Fig. 19. From Fig. 19(b), it can be observed that the RF method produces many false alarms on the sea surface and performs worst in terms of the integrity and boundary precision of the oil slick region, with numerous undetected areas within the oil slicks. As shown in Fig. 19(c), compared to the RF method, Improved-2D-CNN shows better integrity in the oil slick region, but more noise appears on the sea surface. As illustrated in Fig. 19(d) and Fig. 19(e),

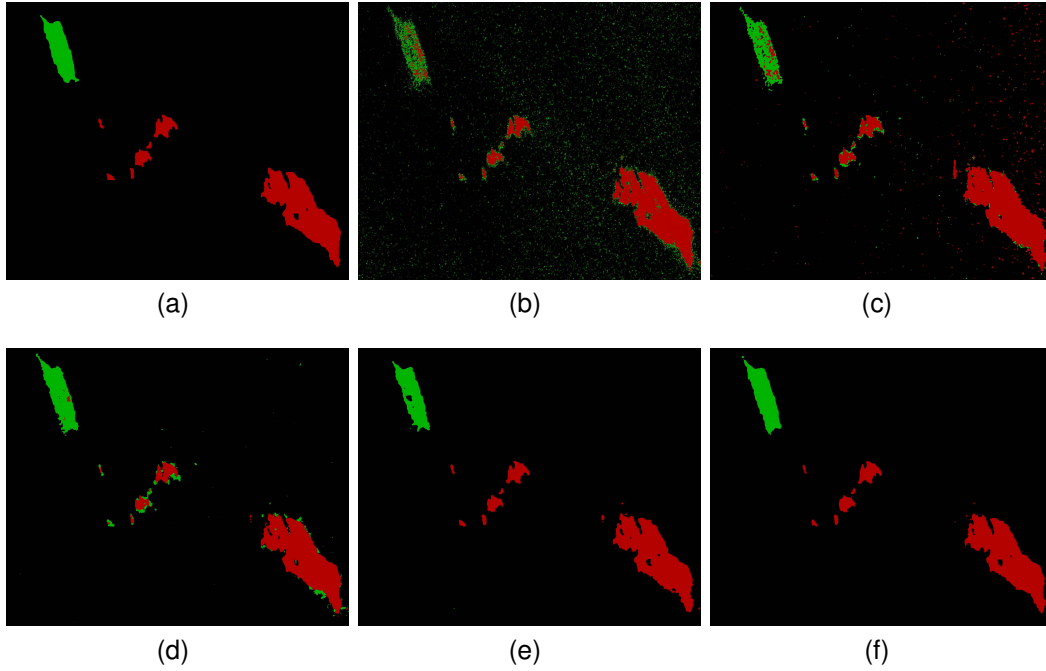


Fig. 18. The enlarged detection results of the oil spill area on the European North Sea oil spill image. (a) Groud Truth, (b) RF, (c) Improved-2D-CNN, (d) Dual-EndNet, (e) DRSNet, (f) MSHFFNet.

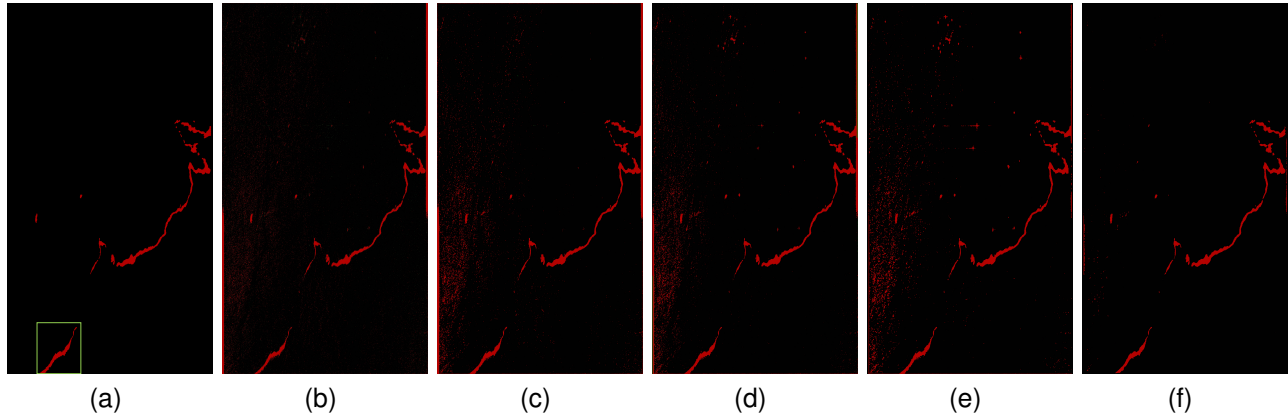


Fig. 19. Results of different methods for Gulf of Mexico oil spill image I. (a) Groud Truth, (b) RF, (c) Improved-2D-CNN, (d) Dual-EndNet, (e) DRSNet, (f) MSHFFNet.

the Dual-endNet and DRSNet methods mistakenly identify oil platforms as oil slicks, which may be due to the inability of the input network's polarimetric features to adequately distinguish between the scattering mechanisms of oil platforms and oil slicks, causing these methods to ineffectively differentiate between oil platforms and crude oil. In Fig. 19(f), our method maintains the integrity of narrow oil regions and exhibits the best performance in suppressing noise, demonstrating that our model is more adept at leveraging contextual information and accurately reconstructing elongated oil slick regions while suppressing interference.

Table VI provides the detection accuracy of different methods for each category, as well as  $OA$ ,  $AA$ ,  $F1$  and  $mIoU$  for image I of the Gulf of Mexico oil spill. Our method exceeds RF, Improved-2D-CNN, Dual-endNet and DRSNet in

classification accuracy for both oil slicks and seawater regions. Our method's  $OA$  is higher by 2.58%, 2.53%, 2.06% and 1.57% compared to RF, Improved-2D-CNN, Dual-endNet and DRSNet respectively. Additionally, our method's  $AA$  shows improvements of 5.86%, 0.92%, 1.27% and 0.63% over these methods. With the highest  $F1$  score, our approach proves more robust, effectively identifying the majority of the oil spill regions while substantially lowering false positives. A  $mIoU$  of 91.37 further validates the superior detection capability of our method concerning oil spill areas.

To better demonstrate the precision of our method around the edges of oil slicks, we enlarge the green boxed area in Fig. 19(a). Fig. 20 shows the enlarged detection results of the oil spill region. As shown in Fig. 20(b), Fig. 20(c), Fig. 20(d) and Fig. 20(e), while the RF, Improved-2D-CNN, Dual-



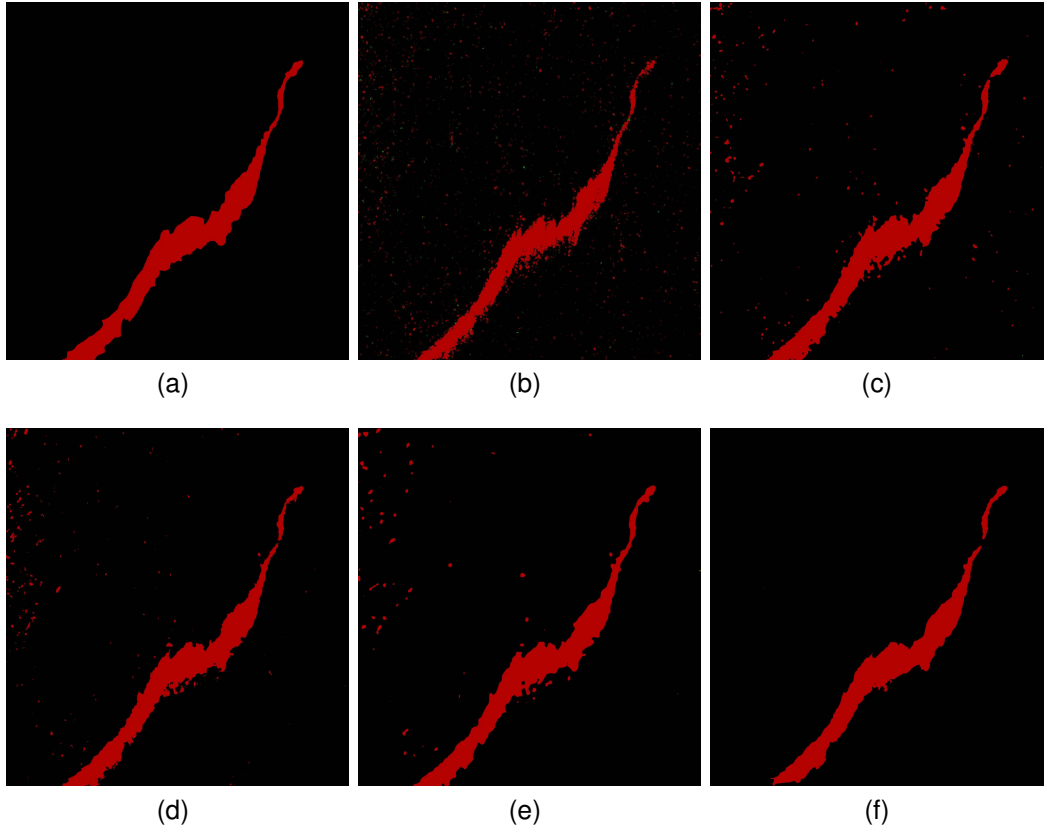


Fig. 20. Enlarge the detection results of the oil spill area on Image I of the Gulf of Mexico oil spill. (a) Groud Truth, (b) RF, (c) Improved-2D-CNN, (d) Dual-EndNet, (e) DRSNet, (f) MSHFFNet.

TABLE VI  
ACCURACY OF METHODS IN GULF OF MEXICO OIL SPILL IMAGE I.

| Method          | Seawater     | Oil          | OA           | AA           | F1          | mIoU         |
|-----------------|--------------|--------------|--------------|--------------|-------------|--------------|
| RF              | 97.35        | 83.96        | 97.18        | 90.65        | 0.47        | 41.55        |
| Improved-2D-CNN | 97.27        | 93.08        | 97.23        | 95.59        | 0.48        | 42.14        |
| Dual-endNet     | 97.76        | 92.71        | 97.70        | 95.24        | 0.50        | 44.06        |
| DRSNet          | 98.25        | 93.11        | 98.19        | 95.88        | 0.51        | 45.63        |
| MSHFFNet        | <b>99.84</b> | <b>93.18</b> | <b>99.76</b> | <b>96.51</b> | <b>0.95</b> | <b>91.37</b> |

endNet and DRSNet methods can detect the general shape of the oil slicks, they exhibit low discriminative power at the oil-water boundaries. As illustrated in Fig. 20(f), our method more closely matches the true shape of the oil region. This approach provides more detailed detection of the oil-water boundaries, retains more detail information, and significantly reduces false alarms at sea, indicating that our method can effectively avoid interference from sea clutter and better restore the shape of oil slicks when detecting elongated oil regions.

### 3) Experimental results for Gulf of Mexico oil spill image II:

To further verify the robustness of the method presented in this paper, we conduct comparative experiments on Image II of the Gulf of Mexico oil spill. The detection results are shown in Fig. 21. In Fig. 21(b), under the RF method, the oil slicks contours in the leeward area are well-detected, but there are missed detections in the windward area, and a significant number of misses along the oil edges. Fig. 21(c) shows the results

from Improved-2D-CNN, which, compared to RF, detects a more complete oil contour in the windward area. However, thin oil film areas are mistakenly identified as seawater. As shown in Fig. 21(d), Dual-EndNet better suppresses sea clutter than RF and Improved-2D-CNN, but it does not effectively detect dispersed oil regions. As depicted in Fig. 21(e), compared to Improved-2D-CNN and Dual-EndNet, the DRSNet method performs better in detecting dispersed oil areas and windward oil slicks, but struggles with detecting small oil patches. In Fig. 21(f), our method shows superior identification and detection results for the oil film. Although there are some local voids inside the oil film in the windward area, possibly due to the oil film being thinner under the influence of the wind, and its backscattering energy being close to that of seawater. However, compared to other methods, our approach detects small-scale oil films and elongated oil films more comprehensively, further demonstrating its effectiveness in detection tasks. Quantitative evaluation metrics are presented in Table. VII. Our method's OA is higher by 0.87%, 0.41% and 0.01% compared to RF, Improved-2D-CNN and DRSNet, respectively. Our method's AA is higher by 3.23%, 0.74%, 3.31% and 2.19% compared to RF, Improved-2D-CNN, Dual-EndNet and DRSNet. High F1 and mIoU scores validate our method's capability to precisely evaluate the extent and configuration of oil spills, illustrating its robustness and reliability in complex detection scenarios.

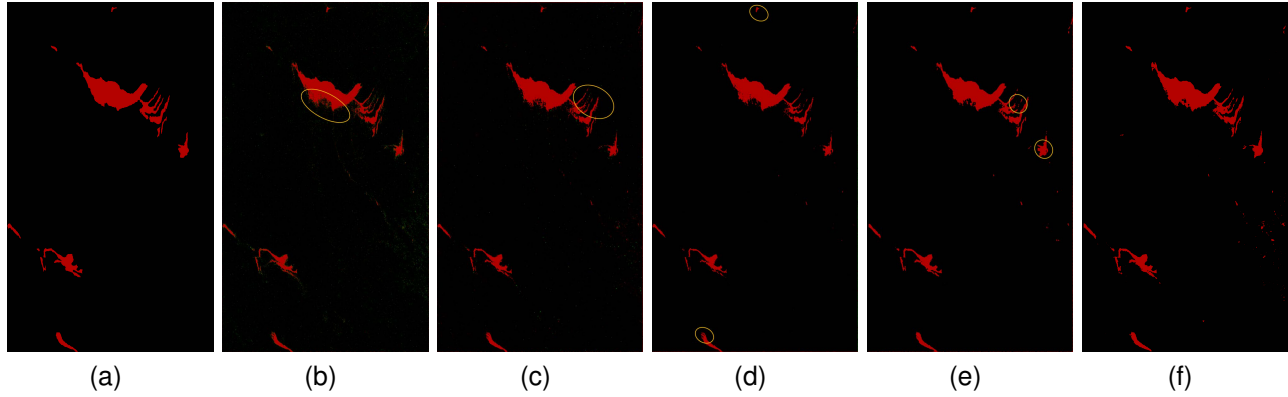


Fig. 21. Results of Different Methods for Gulf of Mexico Oil Spill Image II. (a) Groud Truth, (b) RF, (c) Improved-2D-CNN, (d) Dual-EndNet, (e) DRSNet, (f) MSHFFNet.

TABLE VII  
ACCURACY OF METHODS IN GULF OF MEXICO SPILL IMAGE II.

| Method          | Sea   | Oil          | OA    | AA           | F1          | mIoU         |
|-----------------|-------|--------------|-------|--------------|-------------|--------------|
| RF              | 98.88 | 85.25        | 98.52 | 92.06        | 0.61        | 57.55        |
| Improved-2D-CNN | 99.22 | 89.87        | 98.98 | 94.55        | 0.60        | 56.85        |
| Dual-endNet     | 99.81 | 84.14        | 99.40 | 91.98        | 0.63        | 60.05        |
| DRSNet          | 99.60 | 91.11        | 98.38 | 95.36        | 0.63        | 59.63        |
| MSHFFNet        | 99.49 | <b>95.62</b> | 99.39 | <b>97.55</b> | <b>0.94</b> | <b>89.90</b> |

#### IV. DISCUSSION

##### A. Model effectiveness

To validate the efficacy of the feature hybrid pooling module and feature fusion module within our method's decoder, we conduct three ablation studies, each utilizing the same encoder configuration. In the first experiment, both the feature hybrid pooling module and feature fusion module are omitted, using only the classic pyramid parsing structure of the semantic segmentation network to construct the entire decoder. The second experiment remove the feature fusion module, relying solely on the outputs from the feature hybrid pooling module for upsampling. The third experiment remove the hybrid pooling module and perform upsampling on the features after pyramid parsing and feature fusion. Fig. 22 shows the oil spill detection results of our method and the three ablation studies for the third scene data. From Fig. 22(a), it can be seen that both the feature hybrid pooling module and feature fusion module enhance the detection of oil slicks. As shown in Fig. 22(b), the performance is the worst without two modules. As shown in Fig. 22(c), using only the feature fusion module leads to difficulties in accurately detecting elongated oil slicks. As shown in Fig. 22(d), models using only the feature hybrid pooling module struggle in correctly detecting areas with thin oil films. Therefore, the model utilizing both modules is able to best preserve the details and boundaries of the oil film, achieving the most optimal results.

Detailed quantitative evaluation metrics are presented in Table VIII. Our method's *OA* is higher by 0.19%, 0.36% and 0.14% compared to the other ablation experiments. Our method's *AA* is higher by 6.09%, 3.79% and 3.80% compared to the other ablation experiments. Our method's *mIoU*

reached 89.90, significantly outperforming the other ablation studies. Overall, the feature hybrid pooling module and feature fusion module proposed in this paper effectively enhance the precision of oil-water boundary detection and maximally preserve the detail information of the oil film, demonstrating the method's excellent performance.

TABLE VIII  
DETECTION ACCURACY OF DIFFERENT PARAMETER SETTINGS.

| Net                 | Sea   | Oil          | OA           | AA           | F1          | mIoU         |
|---------------------|-------|--------------|--------------|--------------|-------------|--------------|
| No Fusion & Pooling | 99.65 | 83.27        | 99.22        | 91.46        | 0.92        | 86.44        |
| No Fusion           | 99.34 | 88.19        | 99.05        | 93.76        | 0.91        | 84.95        |
| No Pooling          | 99.57 | 87.93        | 99.27        | 93.75        | 0.90        | 85.15        |
| Proposed            | 99.49 | <b>95.62</b> | <b>99.39</b> | <b>97.55</b> | <b>0.94</b> | <b>89.90</b> |

##### B. Feature effectiveness

To validate the effectiveness of the feature proposed in this paper, we design three ablation experiments. In the first experiment, only polarimetric scattering intensity information is used as the input for MSHFFNet. In the second experiment,  $H$  and polarimetric scattering intensity information are used as the input for MSHFFNet. In the third experiment,  $H_p$  and polarimetric scattering intensity information are used as the input for MSHFFNet. To better observe the advantages of the feature, Fig. 23 presents the oil spill detection results of the proposed method and the three ablation experiments on the enlarged first scene data. As shown in Fig. 23(a), using  $H_c$  and polarimetric scattering intensity information as network input more effectively distinguishes between oil spills and interference. The performance is worst when only polarimetric scattering intensity information is used. In the results using  $H$  and polarimetric scattering intensity information, the crude oil in the lower right corner is incorrectly classified as plant oil. In the results using  $H_p$  and polarimetric scattering intensity information, the edges of the crude oil are misclassified as plant oil. Therefore, using  $H_c$  and polarimetric scattering intensity information as the input for MSHFFNet achieves better results.

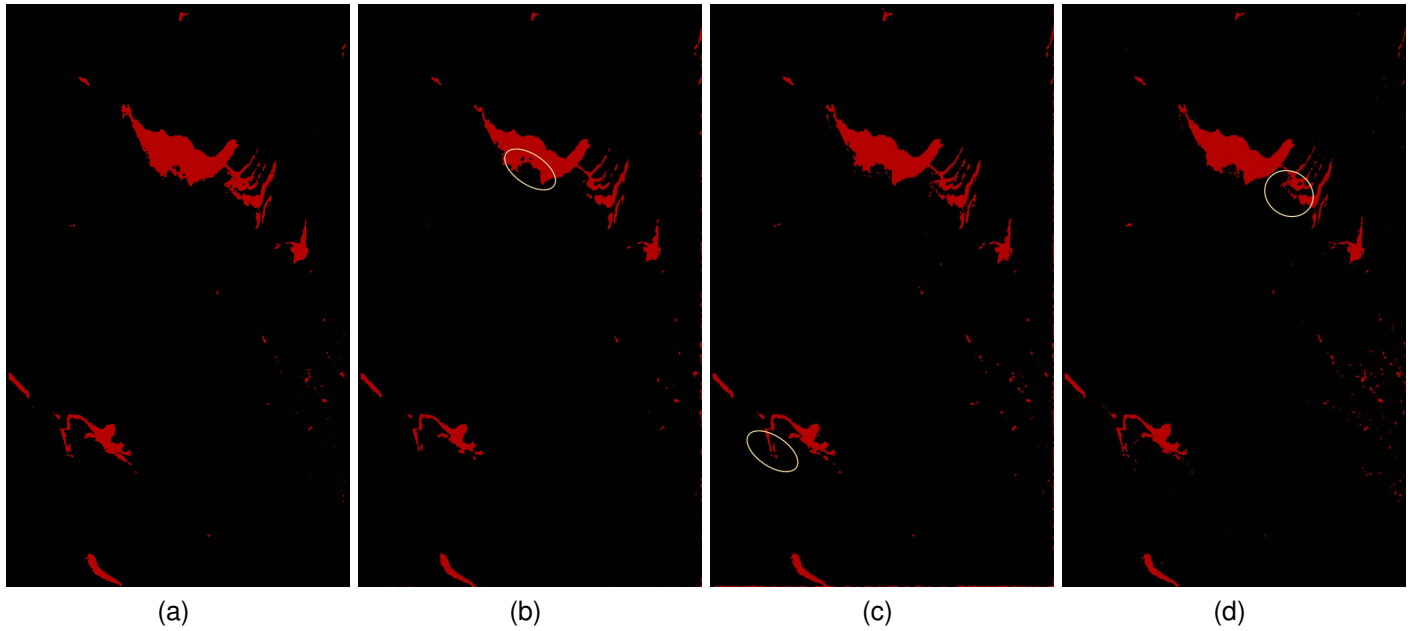


Fig. 22. Detection results with different parameter settings. (a) Proposed, (b) No Fusion & Pooling, (c) No Pooling, (d) No Fusion.

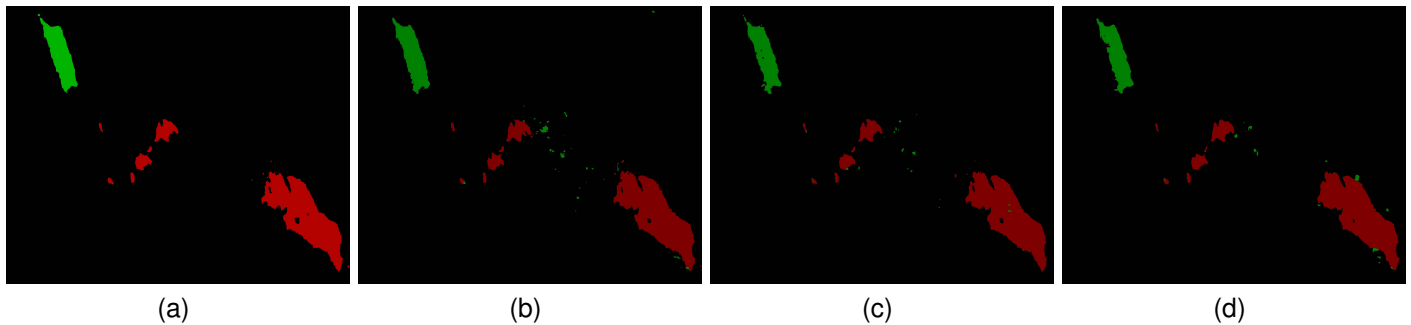


Fig. 23. Detection results with different feature settings. (a) Proposed, (b)  $T_{11}$ ,  $T_{22}$ ,  $T_{33}$ , (c)  $T_{11}$ ,  $T_{22}$ ,  $T_{33}$ ,  $H$  (d)  $T_{11}$ ,  $T_{22}$ ,  $T_{33}$ ,  $H_p$

TABLE IX  
DETECTION ACCURACY OF DIFFERENT FEATURE SETTINGS.

| Method                                 | Seawater     | Plant oil | Crude oil    | $OA$         | $AA$         | $F1$        | $mIoU$       |
|----------------------------------------|--------------|-----------|--------------|--------------|--------------|-------------|--------------|
| $T_{11}$ , $T_{22}$ , $T_{33}$         | 99.19        | 96.77     | 90.52        | 99.17        | 95.50        | 0.83        | 74.78        |
| $T_{11}$ , $T_{22}$ , $T_{33}$ , $H$   | 99.88        | 96.81     | 90.16        | 99.85        | 95.62        | 0.90        | 84.38        |
| $T_{11}$ , $T_{22}$ , $T_{33}$ , $H_p$ | 99.87        | 96.91     | 90.10        | 99.83        | 95.63        | 0.89        | 83.39        |
| Proposed                               | <b>99.96</b> | 96.52     | <b>97.40</b> | <b>99.94</b> | <b>97.96</b> | <b>0.97</b> | <b>94.04</b> |

For detailed quantitative evaluation metrics, please refer to Table IX. Using  $H_c$  and polarimetric scattering intensity information as the network input achieves the highest classification accuracy for oil types, with high F1 scores and mIoU values collectively confirming that the feature effectively determines the extent of the oil spill while avoiding the influence of interference.

### C. Noise analysis

When utilizing polarimetric SAR for oil spill detection, the system's noise equivalent sigma zero (NESZ) has a certain degree of impact on the image's oil spill detection capability.

When the backscattered power of different polarimetric channels approach or fall below the NESZ, they are considered to be corrupted by noise pollution. To investigate whether the changes in the values of features in this method are influenced by low signal-to-noise ratios (SNR), this paper conducts a SNR level analysis based on two scenes of Radarsat-2 images. The selected samples from the two images are shown in Fig. 1(b) and Fig. 3.

As shown in Fig. 24, the normalized radar cross section (NRCS) of co-polarized channels (VV, HH) and cross-polarized channels (VH) was plotted using NESZ. Fig. 24(a)–24(c) present the noise analysis for different polarimetric channels under three sample sets: crude oil, plant oil, and



seawater from the European North Sea oil spill image. In the VV channel, all three samples are above the NESZ baseline. The HH channel shows a similar pattern to the VV channel, with all samples above the NESZ. However, the signal levels for the three sample types are slightly lower than those in the VV channel. In the VH channel, all data samples fluctuate near the NESZ baseline, indicating that the cross-polarized channel is highly affected by noise pollution.

Fig. 24(d)–24(f) show the noise analysis of the Gulf of Mexico oil spill image I for oil slicks and seawater in different polarimetric channels. In the VV channel, the seawater sample data are above the NESZ baseline, and most of the oil slick samples are also above the NESZ baseline. Compared to the VV channel, in the HH channel, most of the data samples fall below the NESZ baseline in the VH channel, all of them are subject to noise pollution, and the data distribution overlaps significantly, making it indistinguishable. Additionally, under the same sensor, the NESZ baseline for the Gulf of Mexico oil spill image I is higher than that for the European North Sea oil spill image, possibly due to a larger incident angle in the Gulf of Mexico oil spill image I [19].

The results from the two scenes indicate that co-polarized channels provide a higher signal-to-noise ratio than cross-polarized channels, with the cross-polarized channel being most affected by noise pollution. Previous studies have suggested that a high entropy value of the feature could be due to multiple scattering as well as low SNR. [36], [43], [44] This conclusion is validated in this paper. In future research, we will further analyze the impact of SNR on features.

## V. CONCLUSION

In this paper, we introduce a multi-scale hybrid feature fusion network that combines composite polarimetric scattering entropy  $H_c$  and polarimetric intensity information from  $T_{11}$ ,  $T_{22}$  and  $T_{33}$  utilizing fully polarimetric SAR data for precise oil spill detection. The network distinguishes between oil slicks and look-alikes and enhances the detection accuracy of oil-water boundaries, effectively addressing existing challenges. Initially, we propose composite polarimetric scattering entropy  $H_c$ , developed by integrating polarimetric entropy and Freeman-based entropy, tailored to the differential scattering characteristics between oil spills, look-alikes and seawater, significantly enhancing the contrast between oil slicks and its analogues. Through both qualitative and quantitative analyses,  $H_c$  demonstrates clear advantages over traditional polarimetric features in detecting oil spills. Subsequently,  $H_c$ , along with  $T_{11}$ ,  $T_{22}$  and  $T_{33}$ , serves as inputs to MSHFFNet. Within the network's encoder, Resnet34 is utilized for feature extraction. The decoder incorporates a hybrid pooling module and a feature fusion module. The hybrid pooling module effectively captures both short-range and long-range dependencies among features of different scales. Meanwhile, the feature fusion module leverages sSE blocks to weight and merge shallow and deep features, ensuring that both detailed information and semantic information are captured while focusing on small-scale oil slick regions. The method was validated using three types of oil spill data from the Radarsat-2 sensor. Comparative

results reveal that MSHFFNet substantially surpasses competing methodologies in key performance metrics such as  $mIoU$  and category accuracy. It significantly improves the detection accuracy of elongated and small-scale oil slicks while minimizing false positives caused by sea clutter or biogenic oils. Future work will explore how to further optimize and adjust the network model under complex conditions such as different sensors, oil film thicknesses, and high wind scenarios to enhance the accuracy and robustness of oil spill detection.

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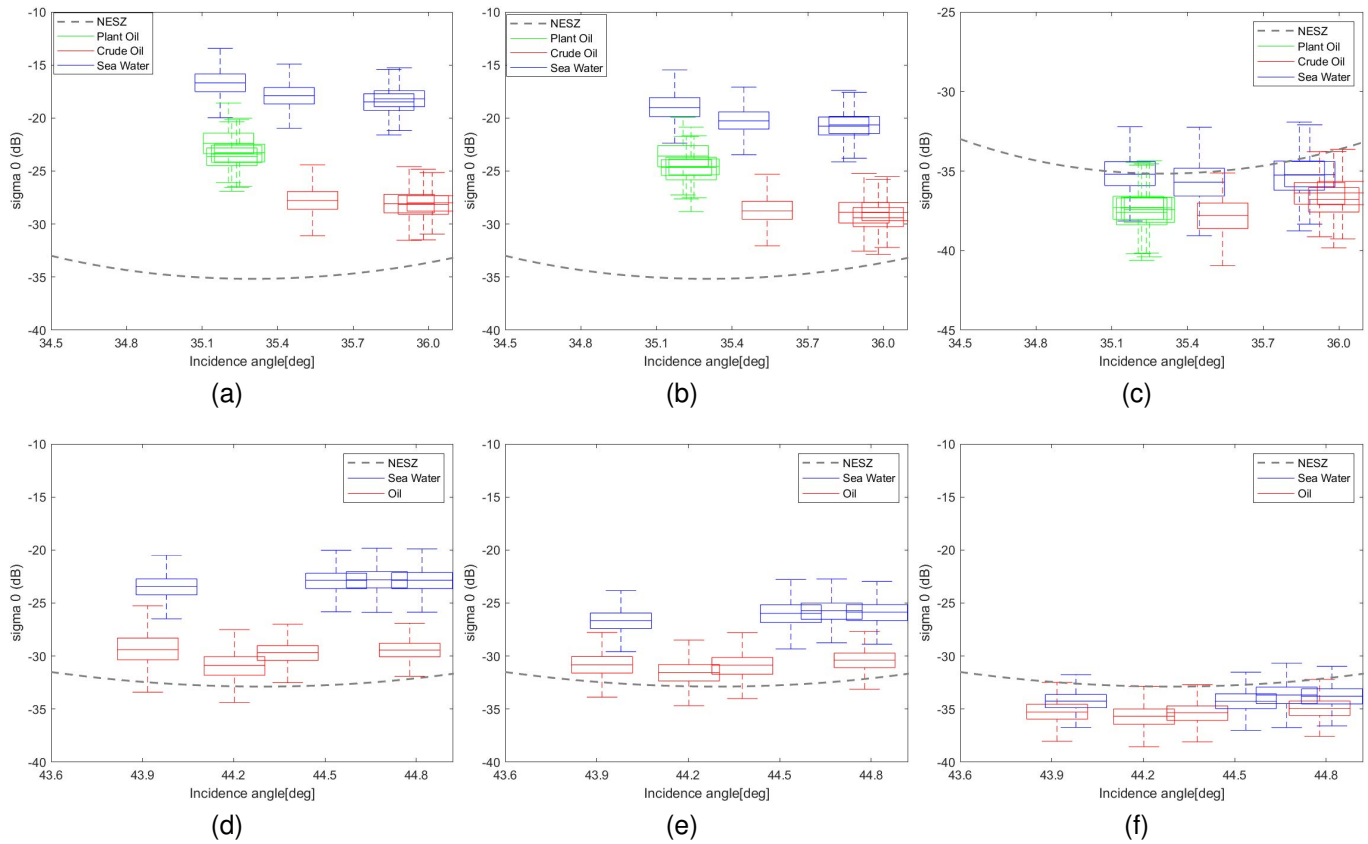


Fig. 24. Signal-to-noise analyses of HH, VV and VH channels (a) European North Sea-VV, (b) European North Sea-HH, (c) European North Sea-VH (d) Gulf of Mexico-VV, (e) Gulf of Mexico-HH, (f) Gulf of Mexico-VH.

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